SCALAR SOURCES AND UNIFORM INSERTIONS FOR LATTICE YANG–MILLS FIELDS

Abstract. We develop scalar action-density sources on admitted lattice gauge trajectories, with the real reference construction held fixed. Local source formation and retained-integral estimates give holomorphic source energies and a complex partition upper bound. Reflection and chessboard estimates transfer these bounds to every periodic Wilson Gibbs state at fixed cutoff, using a common cell normalizer independent of the state. Rooted domain counting yields graded separated moments. A separate argument on the reflected observable algebra gives a one-field insertion bound independent of the surrounding polynomial degrees and number of fields. The local analytic formation and history bounds remain explicit inputs. Conditional passage of the insertion estimate to convergent separated moments is included; existence of a continuum limit is not asserted.

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1. Introduction

A spatially varying source for the Wilson action must pass through the same local transformations as the original measure. At complex source values positivity is unavailable. The source dependence of the local interactions, their extracted coefficients and their numerical vacuum terms must therefore be controlled before summing the histories. An unsourced partition bound does not provide that control. We keep the real reference data fixed, identify the exact source transport and local scalar subtractions, and deduce the sourced upper bound from the stated local analytic and integral estimates.

The passage to infinite volume uses a different argument. Reflection and chessboard domination compare every periodic Gibbs state with finite tori at fixed lattice spacing. The holomorphic normalizer used in this comparison is selected from the finite-torus local energies before a state is chosen. Its first derivative may grow with the cutoff, but it cancels when comparing state means or centering observables at another state. This permits estimates for all such states without treating each state as a torus limit.

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There are two distinct conclusions for products of fields. A rooted domain count and a choice of cell offsets give a product source polydisc for separated pieces. Cauchy's formula then yields one fixed factorial order for moments and cumulants. To control an insertion in the reflected norm of an arbitrary polynomial, we instead repeat a cell insertion along a time column. Log-convexity of its raw reflection moments removes the endpoint constants. Boundedness on the Hilbert quotient is a consequence of this argument, rather than a premise of it.

The localization and connected expansion methods have their antecedents in Bałaban's small-field construction [5, §§1–2, pp. 2–22]. The present deductions also use the finite reflection and Gibbs-state results of [2], the local block conventions of [1], the regular-family interfaces of [3], and the admitted real trajectories and unsourced floor of [4]. The actual nonlinear source domains, local formation and retained mass estimates are named hypotheses at the points where they enter. The conclusions do not include an interacting two-point asymptotic law or a construction of continuum fields.

2. Finite sources and fixed reference data

Fix an odd integer $L \geq 3$ and a cutoff $K \geq 0$. Put $a = L^{-K}$. Fix also a nonnegative integer m_0 and put $M = L^{m_0}$; localization cubes have side M in their own mesh units. On the periodic lattice $a\mathbb{Z}^4/\ell\mathbb{Z}^4$, assign $U_b \in \mathrm{SU}(2)$ to every positive bond and set $U_{\bar{b}} = U_b^{-1}$. All periods and localization partitions used below are required to be compatible. Each unoriented plaquette is counted once, with a prescribed orientation and barycentre $x(p)$. Write

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{Tr} U_p, \qquad A(U) = \sum_p a_p(U), \qquad \mathcal{O}_a(f) = \sum_p f(x(p))a_p(U). \tag{2.i}$$

Here $0 \leq a_p \leq 2$ on real configurations. There is no additional factor a^4 in the definition of $\mathcal{O}_a$.

The half trace fixes the smooth-field normalization. For a Hermitian traceless matrix F use $|F|^2 = \frac{1}{2}\operatorname{Tr} F^2$. If on a fixed compact region the elementary holonomies of a smooth connection have logarithms $ia^2F_{\mu\nu}(x(p)) + O(a^3)$ with uniform remainder, these small logarithms are traceless. The quadratic term gives

$$a_p(U) = \tfrac{1}{2}a^4|F_{\mu\nu}(x(p))|^2 + O(a^5).$$

For a fixed continuous test f supported in this region, the weighted sum in eq. 2.i is a Riemann sum for $\frac{1}{2}\int f \sum_{\mu<\nu}|F_{\mu\nu}|^2$. The estimates below apply to the lattice Wilson measure through the finite observable in eq. 2.i.

Under a site gauge the plaquette product is conjugated at its base point, so its trace and every $\mathcal{O}_a(f)$ are gauge invariant. Reversing a plaquette orientation replaces its matrix by its inverse, which has the same trace in SU(2). A lattice translation or signed coordinate permutation carries the observable to the one with the correspondingly transported barycentric test. Reindexing the bonds preserves product Haar measure and the Wilson action, so finite-volume expectations have this covariance whenever the transformation preserves the torus. With Haar probability measure on each bond, set

$$Z_{a,\ell,\beta}(W) = \int \exp\{-\beta A(U) + \mathcal{O}_a(W)\}\prod_b dU_b, \qquad \beta > 0.$$

The unsourced probability law is denoted by $\mu_{a,\ell,\beta}$. Its denominator $Z_{a,\ell,\beta}(0)$ is strictly positive.

The source data are a finite triple $w = (c, (f_i), (z_i))$, whose function is $W = c + \sum_i z_i f_i$. Here $c, z_i \in \mathbb{C}$ and the f_i are real Lipschitz torus functions; the family may be empty. Fix the geometric constant $c_M > 0$. For $s \geq 0$, put

$$\|f\|_{\sharp,s} = \max\{\|f\|_\infty, c_M L^{-s}\operatorname{Lip}(f)\}, \qquad \|w\|_{\flat,s} = |c| + \sum_i |z_i|\|f_i\|_{\sharp,s}. \tag{2.ii}$$

At $s = 0$ omit the second subscript. This is the physical source size used in the volume-limit estimates. For real C^1 functions the Lipschitz seminorm can be replaced by the supremum of the Euclidean gradient norm. For a complex Lipschitz test use the same $\sharp, s$ formula, with the Lipschitz seminorm of its map into $\mathbb{C}$. Its real and imaginary parts each have norm at most that of the test, so their two-profile representation costs at most twice that norm. The number $\|w\|_{\flat,s}$ belongs to the chosen triple; it can change when the same function is represented by another list. It bounds both $\|W\|_\infty$ and $c_M L^{-s} \operatorname{Lip}(W)$. Every local source term must agree for two representations with the same values on all the source points that it reads. Holomorphy means holomorphy in (c, z) for each fixed real family, with these compatible values. The ordinary source class consists of all triples with $\|w\|_\flat < 2r$, for one fixed $r > 0$.

Write $a_s = L^{-s}$ for physical depth scales. If $m = K - s$ and $j \le m$, the source size gives the exact freezing estimate

$$c_M a_{K-j} \operatorname{Lip}(W) = L^{-(m-j)} c_M a_s \operatorname{Lip}(W) \le L^{-(m-j)} \|w\|_{\flat,s}. \tag{2.iii}$$

This gain concerns variation of the same source function across an earlier physical block.

The real functions permitted in this ball include global torus profiles obtained by reflection of a function supported inside one cell. They need not have support in a single ball. This inclusion will allow the chessboard construction to use one real profile for all even copies and another for all odd copies.

Lemma 2.1 (Finite source derivatives)**.** For a fixed finite torus and real bounded tests $f_1, \dots, f_n$, the function $Z(\sum_i z_i f_i)$ is entire in $z \in \mathbb{C}^n$. Its mixed derivatives are obtained by inserting the corresponding products of $\mathcal{O}_a(f_i)$ in the integral. A holomorphic logarithm of $Z(z)/Z(0)$ exists near zero. Its mixed derivative, with one derivative in each variable, is

$$\kappa_n^\mu(\mathcal{O}_a(f_1), \dots, \mathcal{O}_a(f_n)) = \sum_\pi (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \mu\left(\prod_{i \in B} \mathcal{O}_a(f_i) \right), \tag{2.iv}$$

where π ranges over the set partitions of $\{1, \dots, n\}$. For $n \ge 2$ this value is unchanged if any test observable is shifted by a constant. Conversely, each product moment is the sum over set partitions of the products of the corresponding block cumulants. ■

Proof. Each observable is bounded at this fixed cutoff and volume. On a compact parameter set, its exponential and every fixed derivative have one integrable constant bound times the positive Wilson density. Dominated convergence proves differentiation under the integral and holomorphy. Continuity and $Z(0) > 0$ give a neighborhood on which $Z(z)/Z(0)$ lies in the disc of radius $1/2$ about one. The logarithm series is holomorphic there.

Put $F(z) = Z(z)/Z(0)$. In $\log F = \sum_{k \ge 1} (-1)^{k-1} (F - 1)^k / k$, the coefficient using each of the n variables once receives contributions only from $k \le n$. Assigning variables to the k nonempty ordered factors amounts to ordering the blocks of a set partition. The $k!$ orderings and the factor $1/k$ give $(k - 1)!$, proving eq. 2.iv. Expanding $F = \exp(\log F)$ gives the inverse formula: each ordering of the k blocks is cancelled by the exponential coefficient $1/k!$. Shifting an observable multiplies F by the exponential of a linear function, which changes only the first derivatives of its logarithm. □

We use an admitted real reference run from the trajectory construction [4]. Its chosen weights x_j, reference weights θ_j, cutoffs, partitions, gauges and positive history kernels are frozen when a source is introduced. The reference Gaussian is the one of that real run. A complex source changes the inserted density; it supplies no positivity for a new complex Wilson law. The local analytic formation and retained-integral bounds assumed below concern this fixed real construction.

For estimates on infinite lattices, a periodic Wilson Gibbs state means a probability measure invariant under a finite-index group of lattice translations and satisfying the Wilson DLR specification at the same fixed a and β. We use the reflection, chessboard and all-state domination

results of [2]. In particular [2, Theorem B] gives invariance under every lattice translation for these states. The volume limit is taken at fixed a and β. Statements comparing two states use the same cutoff and coupling for both. Existence of a continuum limit is not an input to these finite-spacing arguments.

Constants denoted by C may increase between estimates. Unless a dependence is displayed or stated, they depend only on the fixed geometric and analytic input bounds, not on the cutoff, compatible volume, source representation or Gibbs state. Constants indexed by the number of observables may depend on that number. Fixed-cutoff bounds for point counterterms and for normalizers before taking a volume subsequence are explicitly distinguished from uniform bounds.

3. Source transport with fixed reference data

At level j the mesh is $h_j = L^{j-K}$. A source is evaluated at the bare plaquette barycentres, including the plaquette points used as anchors by earlier levels. Fix the reference choices before assigning a source. These choices include the background and current maps, real Gaussian precision and gauge weight, every test and guard, and the reference laws used when an interior field is replaced. The sourced Wilson weight is $e^{-A(\beta-W,U)}$, where $A(u,U)=\sum_p u(p)a_p(U)$. In particular a constant real source v changes the bare weight to $\beta - v$ in this same reference.

Proposition 3.1 (Transport of the bare source)**.** Use a finite admitted prefix of the fixed reference run. Let $K_h(U, dV)$ be its positive history kernels through any level j, with $\sum_h K_h(U, \mathcal{V}_j) = 1$. Then the sourced terms are

$$\tau_h^W(dV) = \int e^{-\beta A(U)} e^{A(W,U)} K_h(U, dV)\, dU, \qquad \sum_h \tau_h^W(\mathcal{V}_j) = Z(W). \tag{3.i}$$

On each finite source family the measures τ_h^W are entire in total variation. For a real source they are positive measures. The same formula transports a bounded Borel gauge-invariant insertion $F(U)$ by putting F in the integrand. ■

Proof. The history kernels are formed by the normalized guarded averages, gauge draws, coordinate changes and reference replacements of the finite run [4]. Each actual move acts linearly on the incoming measure, and the outgoing kernels sum to a probability kernel. A gauge draw uses its normalized kernel, with the orbit representatives and every test retained in their order. Integrating the draw against its input gives the gauge-weight formula when that input is invariant under the gauge subgroup of the draw. For this identification use the actual retained-gauge equivariance at each draw, which preserves invariance of the weighted incoming term under that draw subgroup. The bare Wilson source and the allowed insertion are gauge invariant. Composition and integration give eq. 3.i. On bounded sets of source coefficients the exponential series converges uniformly on the compact bare configuration space, since $0 \le a_p \le 2$ and there are finitely many plaquettes. Integration against any fixed finite history kernel is bounded in total variation. This proves the entire assertion and the same argument gives the insertion formula. For real profiles the integrand defining the sourced measure is nonnegative. The partition of outcomes depends on the fields and the fixed tests, so no outcome is introduced by complexification. □

Proposition 3.2 (Block observables outside replaced interiors)**.** Suppose the actual averaged stage sends each incoming history term under $\mathcal{M}$ to the sum of its tested continuations. Suppose also that for each tested continuation the replacement stage preserves the marginal on bonds exterior to the designated replacement components. It suffices that these identities hold on gauge-invariant bounded Borel tests, including all such tests of the exterior bonds in the second identity. Use the covariant local block map of [1]. At $W = 0$ there is a joint probability law of U, h, V whose bare marginal is the Wilson probability and whose history/block marginal is $Z(0)^{-1} \sum_h \delta_h \otimes \overline{\tau_h^0}$, where the bar is full site-gauge averaging. The law is invariant under simultaneous bare gauges and their restrictions on V. On a level-j bond whose complete backward block read misses the enlarged components acted on along h, it has $V(c) = \mathcal{M}^j U(c)$ almost surely.

For bounded gauge-invariant F, let $F_h(V)$ be the invariant version of the conditional expectation of $F(\mathcal{M}^j U)$ given (h, V). Then

$$\mu(F \circ \mathcal{M}^j) = Z(0)^{-1} \sum_h \int F_h \, d\tau_h^0, \qquad |F_h| \le \|F\|_\infty \quad \tau_h^0\text{-almost everywhere.} \tag{3.ii}$$

If F reads only the bonds just specified for h, then $F_h = F$ τ_h^0-almost everywhere. Omitting any event of histories in eq. 3.ii costs at most its probability times $\|F\|_\infty$. ■

Proof. Average the incoming, tested and completed terms under the full site gauge groups. The identities on invariant tests become measure identities, since any bounded test of an invariant measure has the same integral as its Haar average. Projection to the exterior commutes with gauge averaging. The block map commutes with the restriction of a site gauge, and restriction carries Haar probability to Haar probability on the coarser sites.

Fix an incoming history h. If λ is its averaged term and λ_b the averaged tested terms for its continuations, then $\sum_b \lambda_b = \mathcal{M}_* \lambda$. Write $r_b = d\lambda_b / d(\mathcal{M}_* \lambda)$. From V with measure λ choose b with probability $r_b(\mathcal{M}V)$ and retain $\mathcal{M}V$ as the tested field. For each b the sum of completed terms, with their continuation labels, has the same exterior marginal as λ_b. Disintegrate this sum over the exterior field and use its conditional probability as the second draw. The field after this draw agrees with the tested field on the exterior. All configuration spaces are finite products of compact groups and all label sets are finite, so the conditional laws exist. Iterating these two draws gives the required marginals. At each step the complete block read includes the fine plaquettes for its branch tests and all chosen contours, including projected short seam reads. It lies in the projected double block by the locality result of [1]. Under nested centred blocks all its recursive reads lie in the backward double block. Consequently, if the complete backward read misses all the acted-on enlarged components, induction identifies every read with the corresponding read of the bare iterated map. The final read therefore agrees. Averaging the joint law under the simultaneous gauge action preserves both marginals and these equality events.

The conditional expectation can be averaged under site gauges on V. Invariance of the joint law and surjectivity of gauge restriction make the average a version of the same conditional expectation. Its essential bounds and its equality to F on the specified histories concern invariant events and so pass from $\overline{\tau_h^0}$ to τ_h^0. Integrating the conditional expectation proves eq. 3.ii. Positivity of the omitted measures proves the last bound. In particular a supplied one-cell rarity bound gives the corresponding finite union bound for a list of cells. □

4. The running scalar weight

On each fixed formation window, let $h_y^{(j)}(p)$ be the nonnegative tents of the step with $\sum_y h_y^{(j)}(p) = 1$. The anchor p_y is a fixed bare plaquette and $h_y^{(j)}(p) \neq 0$ implies $d(x(p), x(p_y)) \le CMh_j$. Let $0 \le \varphi_j(p) \le 1$ be the actual stage profile. Write $G_{j,y}(v)$ for the constant-source whole-lattice member used at y, with the fixed read and pointing of the step. It includes the companion needed for its physical quadratic readout. For constant v the member has the covariance required by the

pointed coefficient theorem of [3]. At the trivial pair $P_\circ = (\mathbf{1}, 0)$ subtract its vacuum value, and use its own coefficient $b_j(v)$ to subtract its local action. In the unit-Wilson convention this is the operation

$$G_{j,y}(v) - G_{j,y}(v)(P_\circ) - b_j(v)A(h_y^{(j)}, \cdot). \tag{4.i}$$

Here the action denotes the actual pointed classical member, with its exact restriction and inherited transport. At a site where the array selects $G_{j,y}(W(p_y))$, the coefficient in this formula is $b_j(W(p_y))$. The coefficient cancels the quadratic moment of that member by the linear readout in [3]. There is no requirement that the array of site-dependent members be translation invariant. The Euclidean members are supplied on the common reference and common formation domains. When read on an admitted torus they use those same coefficients. A different torus member with no common restriction would require its own readout.

The complete old pointed member and its tent action use a single window read and a single bracket functional at fixed j, y. Linearity allows the bracket to be applied to their vacuum- and quadratic-subtracted sum before estimating it. For an infinite old domain sum, normal convergence on the switch disc justifies this identity by Cauchy's formula. Truncation to near domains leaves the omitted part in the separate far-tail estimate. The coefficient is attached to the complete old pointed member. It is neither an individual domain coefficient nor a claim about the readout of the new bracket output. Differentiating the fixed linear readout and subtraction in eq. 4.i gives the same subtraction for the source derivative with its own coefficient.

Proposition 4.1 (Scalar shifts and their source range)**.** Suppose the actual constant-source formation is available at each step on the admitted half product of the relative weight discs of [4], with common older reads. At step i let the nonconstant unmarked coefficient be holomorphic on the full relative product, with bound $C\theta_{i-1}^{-1/2}(\log\theta_{i-1})^{\mathfrak{p}_0}$, and let the formed marked coefficient have bound $C\theta_{i-1}^{-p_R}$ on $|v| < 4r$, where $p_R > 1$. The marked hypothesis is for the actual holomorphic sourced term including inherited remainders. Both coefficient maps are real on real data. Assume the real deviations at the used levels lie in the quarter products and that the relative half products have at least $6r$ of room beyond them. Suppose

$$\theta_j \geq H, \qquad \theta_i - \theta_j \geq \lambda(j-i) - C_0 \quad (i \leq j), \qquad \lambda > 0.$$

For H sufficiently large at fixed r, the successive coefficients on $|v| < 4r$ have radial bounds

$$d_i = \sup_{0<|v|<4r} \frac{|b_i(v) - b_i(0)|}{|v|}, \qquad \sum_{i\leq j} d_i \leq C\{\theta_j^{-1/2}(\log\theta_j)^{\mathfrak{p}_0} + r^{-1}\theta_j^{1-p_R}\} \leq \tfrac{1}{6}. \tag{4.ii}$$

The bound is uniform in admitted volume and prefix length. For the ordinary variable source define

$$\begin{aligned} c_i(p; W) &= \varphi_i(p) \sum_y h_y^{(i)}(p) b_i(W(p_y)), \\ \zeta_j(p; W) &= W(p) + \sum_{i\leq j} \{c_i(p; W) - b_i(0)\varphi_i(p)\}, \\ x_j(p) - \zeta_j(p; W) &= \beta - W(p) - \sum_{i\leq j} c_i(p; W). \end{aligned} \tag{4.iii}$$

This is the exact Wilson weight after the local action subtractions. Each value is holomorphic in the source coefficients and depends only on bare source values within distance CMh_j of p. In particular $|\zeta_j(p; W)| \leq (4/3)\|W\|_\infty$. At a constant real source $|v| < r$ the uniform interior scalar shift satisfies $2/3 \leq \zeta_j'(v) \leq 4/3$, and at zero its derivative differs from one by at most the sum in eq. 4.ii. ∎

Proof. Write $z_l = \theta_l - x_l$ in the constant-member reference run. The fixed-reference coefficient is its common one-loop term plus the ordinary analytic coefficient on $(z_l + \zeta_l(v))_{l<i}$ and the actual marked coefficient $q_i(v)$. This is the prescribed regular and marked splitting of the formed coefficient. At

every half product the absolute derivative row bound of the relative-disc estimate in [4] is at most $C\theta_{i-1}^{-3/2}(\log\theta_{i-1})^{\mathfrak{p}_0}$ at step i. Indeed $\theta_l \ge \theta_{i-1} - C_0$ for $l < i$. If the preceding radial sum is at most $1/6$, then $|\zeta_l(v)| \le (4/3)|v| < 6r$, so the entire segment for the next coefficient lies in the half product. The formed marked term $q_i(v)$ satisfies $|q_i(v) - q_i(0)| \le Cr^{-1}\theta_{i-1}^{-p_R}|v|$ by Schwarz's lemma on its actual source disc. The derivative row estimate and integration on the unmarked segment thus bound d_i by the sum of these two envelopes with a fixed constant. Using $\theta_{i-1} \ge \theta_j + \lambda(j-i+1) - C_0$ and comparing with integrals gives eq. 4.ii. The sum over every prefix is at most $1/6$ when H is large enough. Induction starts at $\zeta_0 = v$ and admits each next weight before forming the next coefficient. This proof concerns the analytic coefficient box, with the separate real weight tube imposed on the physical formation.

In eq. 4.iii, expand the real weight $x_j(p) = \beta - \sum_{i\le j} b_i(0)\varphi_i(p)$ and use the partition of unity. At each step the difference of successive shifted weights is exactly $c_i(p; W)$. Linearity of the action therefore compensates eq. 4.i at its own anchor. The distance claim follows from the support of the tents and $h_i \le h_j$ for $i \le j$. Positivity of the tents and eq. 4.ii give the shift bound. For each holomorphic quotient $(b_i(v) - b_i(0))/v$, the bound is d_i on $|v| < 4r$. At $|v| < 2r$, Cauchy's estimate gives $|b_i'(v)| \le 2d_i$. In the constant interior all stage factors are one, and reality gives the stated derivative bounds. At zero use $|b_i'(0)| \le d_i$. □

For an interior scalar weight, let R be either r or $2r$. If $|\theta_j - x_j| + 2R \le \mathtt{t}_j$, the shift bound gives $|\theta_j - x_j + \zeta_j(v)| < \mathtt{t}_j$ for $|v| < R$. This numerical tube inclusion uses the interior weight at its own level; the actual complex formation still uses its stated field and current domains.

For an ordinary source of size less than $2r$, the coefficient derivative bound on that disc also gives, when $h_y^{(j)}(p) \ne 0$,

$$\begin{aligned} |b_j(W(p_y)) - b_j(W(p))| &\le Cd_jMh_j \operatorname{Lip} W \\ &\le Cd_jL^{-(n-j)}\|w\|_{\mathfrak{b},K-n}, \qquad j \le n. \end{aligned}$$

This freezes only the source value in the coefficient. It does not differentiate the stage profiles.

When $x_j(p) \ge x_{\min} > 0$, the modulus of the shifted Wilson weight at a real field is bounded by

$$|e^{-A(x_j-\zeta_j,U)}| \le e^{-(1-4\|W\|_\infty/(3x_{\min}))A(x_j,U)}. \tag{4.iv}$$

This follows plaquette by plaquette from $a_p \ge 0$. It keeps a fixed fraction of the real Wilson gain when the source radius is fixed and $x_{\min}$ is sufficiently large. It does not give a bound for the retained local integrals with their other sourced factors.

Proposition 4.2 (A real shift in the same frame)**.** Suppose a real prefix has weight-tube room $2r$, and fix a real $|v| < r$. Use the same frozen tests and reference laws for the shifted presentation. Assume the complete matching hypotheses of [4, Proposition 7.1] at each formed step: hereditary physical field–current reads, exact common centered regular formation with the omitted earlier remainders read on their actual pairs, the common vacuum convention and cofinal physical coefficient readout. In particular the composed incoming exponent and omitted remainder totals are holomorphic on a common connected complex fluctuation domain containing zero and the real cut. A nonempty full-dimensional real open patch maps their reads into the hereditary windows. Use the common positive continuous density formula in a reference measure with full support there. Assume a bound $r/3$ on each full increment beyond its shared one-loop term. Require initial admissibility at $\beta - v$ and actual local successor admission in this common frame whenever a formed next weight lies in the tube. This includes the geometric, field and retained tests on the cofinal compatible tori. Then the run at $\beta - v$ in this frame is admitted through the same prefix. Its interior weights and coefficients satisfy

$$x_j(\beta - v) = x_j(\beta) - \zeta_j(v), \qquad b_j(\beta - v) = b_j(v). \tag{4.v}$$

If $E_{(j)}$ is the once-counted vacuum at the stop and the source energies are those of eq. 7.i, then

$$E_{(j)}[\beta - v] = E_{(j)}[\beta] + \sum_{i \le j} \sum_{X} e_{K-i,X}(v). \tag{4.vi}$$

■

Proof. At the bare level the two input measures agree. Their actual positive kernels through any common admitted prefix agree, so their history terms, including the all-small-field term, agree. Inductively suppose eq. 4.v holds through level $i - 1$. Formation at the admitted input for step i can be applied before claiming admission of its output. The possible discrepancy in the next weight is bounded by $(4/3)|v| + 2r/3 < 2r$. The second term is the sum of the two errors beyond the common one-loop coefficient. Thus the new entry lies in the tube. The next-formation clause of [4, Proposition 7.1], using the common reads and complex continuation just stipulated, identifies the two coefficients before closure at this new weight. The update then gives the weight equality at i. Invoke the stated successor admission to close the presentation at this entry before using its density. The full all-small-field densities are continuous on the common open identity window and are equal as measures there in a common reference measure with full support. They therefore agree at the identity. The centered activities and actions vanish there, and the fixed gauge and coordinate factors coincide. The value of the exponent at that point is the once-counted vacuum, including its local source differences in the sourced presentation. This proves eq. 4.vi. No termwise equality of all localized activities in the two presentations is needed. □

At a prescribed stop, differentiation along this common admitted real interval gives $\partial_\beta x_j(\beta) = \zeta_j'(0)$, with the reference frame held fixed. The sum in eq. 4.ii bounds its departure from one at that stop, rather than only at the end of a longer trajectory.

5. Ordinary formation with source-dependent members

Fix $q > p_0 \ge 0$ and write

$$\alpha_j = \theta_j^{-1/2}(\log \theta_j)^q, \qquad \delta_j = \theta_j^{-1/2}(\log \theta_j)^{p_0}, \qquad s_j = \delta_j/\alpha_j.$$

The source is inserted into the Wilson potential at the fixed reference. The local Gaussian precision, its normalizer and the retained tests are the same in the sourced and unsourced integrals.

A localization domain X is a connected union of cubes at the specified level. Write $d_j(X)$ for the shortest cube-edge tree within X that visits its cubes. A member $F(X, P, z)$ has a pointing z at a site of X and a pair argument $P = (U, J)$, where complex bond matrices of U lie in $\mathrm{SL}_2(\mathbb{C})$.

The current J is a bond one-form valued in the complexification of the traceless Hermitian colour space, with the factor-i convention of [3]. The pair ranges over the actual complex core of that member, including its declared restrictions and companion, when used. For a pointed family the exponential norm at rate γ is

$$\|F\|_{j,\gamma} = \sup_{X,P,z} e^{\gamma d_j(X)} |F(X,P,z)|.$$

For an unpointed family omit z. For a potential that also has a fluctuation argument take the supremum over its actual allowed fluctuation coordinates. A rooted sum is a sum over the domains through a fixed cube or assigned pointing. We only use such sums with a strict decay reserve above their counting rate.

An ordinary window is a local formation chart at the fixed real reference, on which all vertices use their specified complex cores. Its complete determining read comprises every coordinate, restriction and guard consulted by the operation, including the reads of its old vertices. The unmarked regular family is formed without the marked old remainders. Subtracting it from the full regular family leaves the marked part, with its inherited reads. Retained operations use their own stored variables and local measures. These are the conventions for the actual regular and history operations of [3, 4]. At each step the regular output is formed first. Its constant-source companion selects the coefficient to subtract at each pointing; the resulting weights are then fixed before closing the new retained and remainder terms.

We use the following local analytic inputs. The localized interaction logarithm is formed on the actual Gaussian and switch domains, with a fixed spare slot for a potential of norm less than $w_* > 0$. Its connected series converges normally and has a uniform exponential norm there. The assigned bond pointing has its own local estimate on the same complete reads. The window bounds for inherited members use the actual complex potential and current domains of those members. A calculation at complete physical pairs does not admit independent currents.

Lemma 5.1 (The fixed-reference source vertex)**.** Let η be the bare mesh in the units of a step and $0 < \delta \le \alpha$. Suppose the actual local pair substitution and its decoupled versions have plaquette matrices within $C\alpha\eta^2$ of the identity and the fluctuation changes a plaquette matrix by at most $C\delta\eta^2$. Suppose the assigned physical bracket at an independent pair has the same admitted bounds. Then the Wilson source vertex after subtracting its background value has local norm at most $CM^4\alpha\delta\|W\|_\infty$. In the ordinary local expansion, the unpointed derivative in an inserted potential of norm ϵ has exponential norm at most $C\epsilon$ at the output rate of the connected series. At a complete physical pair its sum is the expectation of the inserted potential in the normalized tested regular integral. The Gaussian normalizer and the scalar ratio at the fixed real weight have zero source derivative. ∎

Proof. For $Q \in \mathrm{SL}_2(\mathbb{C})$, $1 - \frac{1}{2}\operatorname{Tr} Q = \frac{1}{2}\det(I - Q)$. The difference of determinants of two 2×2 matrices is bounded by a constant times their difference in norm times the sum of their norms. Apply this to the two plaquette differences from I. The displacement and $\delta \le \alpha$ bound the second matrix by $C\alpha\eta^2$ even if the radius is given at the first. The change of one Wilson density is at most $C\alpha\delta\eta^4$. There are at most $CM^4\eta^{-4}$ plaquettes in an assigned local cube and its fixed collar. This proves the vertex bound, including complex substitutions. At an independent pair the assertion uses the admitted local bracket in place of its physical specialization.

For a potential direction v with norm $\epsilon > 0$, insert tv in the actual localized expansion on $|t| < w_*/\epsilon$. The derivative of a product of selected Mayer factors is the sum with one factor $e^{V(Y)} - 1$ replaced by $v(Y)e^{V(Y)}$. The corresponding tests and filing domain remain in that term, including when $V(Y) = 0$. The fixed Gaussian domination and normal convergence justify differentiation term by term. If the bound of the local logarithm on this disc is $Be^{-\gamma d(X)}$, Cauchy on $|t| = w_*/(2\epsilon)$ gives $2B\epsilon e^{-\gamma d(X)}/w_*$. A zero direction contributes zero. The same circles bound the norm Taylor remainder by $CB\epsilon^2/w_*^2$ for $\epsilon < w_*/2$. Thus the bounded linear first jet also differentiates an actual

source curve of potentials in this norm. The auxiliary t-disc need not be a physical source disc. The renewed pointing uses the stated rooted pointing estimate, which is separate from the unpointed Cauchy bound.

At a complete physical pair the exponential of the sum of the local logarithms is the tested Gaussian integral $\Xi(t)$. The connected expansion provides its nonzero branch there. Differentiating the integral and this identity divides the inserted integral by $\Xi(0)$, giving the normalized expectation. All Gaussian scalar terms are fixed and therefore differentiate to zero. No division by a full history mass is involved. □

Lemma 5.2 (Rooted localization of the Wilson insertion)**.** For each plaquette p let C_p be its assigned, bounded face-connected base cubes. Use the actual hereditary local response brackets on every set $Y \supset C_p$. In the response formula let s_D be the fine cube switches, and let ν multiply the fluctuation response. Assume the plaquette source evaluation $F_{p,Y}(s,\nu)$ is jointly holomorphic on $|s_D| < e^{\gamma}$, $|\nu| < \rho$, where $\gamma \geq 1$ and $\rho \geq c\alpha/\delta > 1$. Assume the complete pair substitutions are admitted there with $|F_{p,Y}(s,\nu)| \leq C\alpha^2\eta^4|W(p)|$. The base switches are one. The response has the usual restriction property when switches outside Y are zero. A differenced switch in a component disjoint from the base gives zero. Then the filed scalar-source potential has exponential supremum norm at most $CM^4\alpha\delta\|W\|_\infty$ at any fixed rate below $\gamma - 1$. At a physical pair its sum is the actual Wilson difference between the fluctuating and reference fields. ■

Proof. Put $f(s) = F_{p,Y}(s,1) - F_{p,Y}(s,0)$. Schwarz in ν bounds $|f(s)|$ by $2C\rho^{-1}\alpha^2\eta^4|W(p)|$. For the term filed on Y integrate $\prod_{D\in Y\setminus C_p} \partial_{s_D} f$ over the real unit switch box, keeping the base fixed. Cauchy at each real switch value has radius approaching $e^{\gamma}-1$. The absolute bound on this bracket is

$$C\alpha\delta\eta^4|W(p)|e^{-(\gamma-1)|Y\setminus C_p|}.$$

Each switch is differentiated once, so no factorial is present. By the response rooting property every nonzero component meets the base. Connect the base by a bounded tree and add each other cube through its face-adjacent predecessor, once per cube. This gives $d(Y) \leq |Y \setminus C_p| + O(1)$. The number of base plaquettes in a filed union is at most $CM^4\eta^{-4}(1 + d(Y))$. Absorb this last factor in the rate reserve to get the claimed norm. Repeated use of the fundamental theorem of calculus in the switches expresses the all-one response as the sum of these brackets, including the base-only term. Hereditary restriction identifies these summands with the actual local rule. At a physical pair the endpoints $\nu = 1, 0$ are the fluctuating and reference Wilson terms, respectively. This recovers their exact difference. Independent current pairs only use the admitted brackets and need no global physical substitution. □

Lemma 5.3 (First variations at fixed reference)**.** Let $d\lambda$ be a fixed positive sigma-finite measure containing the Gaussian and Borel test factors of a local leg. On a complex disc about $t = 0$, let V_t, L_t be jointly measurable in parameter and field, and holomorphic in t at almost every field. Suppose L_t is locally uniformly bounded in the field supremum norm and e^{V_t} has a common integrable majorant on each compact parameter disc. Suppose the denominator below is nonzero near zero, and set

$$\langle F\rangle_t = \frac{\int F e^{V_t}\, d\lambda}{\int e^{V_t}\, d\lambda}.$$

Dots denote derivatives at zero. Then all three terms on the right below are absolutely defined, and

$$\partial_t \langle L_t\rangle_t|_0 = \langle \dot L\rangle_0 + \langle L_0 \dot V\rangle_0 - \langle L_0\rangle_0 \langle \dot V\rangle_0. \tag{5.i}$$

For the fixed pointed subtraction in eq. 4.i, let G_t be differentiable in the member norm and let b be the fixed bounded linear readout. Its derivative uses its own coefficient $b(\dot G)$. If b^t is differentiable in bounded operator norm and the classical action A^t is differentiable in a common field chart, then

$$\partial_t\big[b^t(G_t)A^t\big]\Big|_0 = \big[\dot b(G_0) + b^0(\dot G)\big]A^0 + b^0(G_0)\dot A \tag{5.ii}$$

on any common differentiable field chart. ■

Proof. Cauchy's formula bounds $\dot L$ in the field supremum norm. The same formula applied to e^{V_t}, with its common integrable majorant, gives an integrable bound for $\dot V e^{V_0}$. Multiplying the latter by bounded L_0 proves the separate absolute integrability claimed in the statement. Cauchy and Fubini also differentiate both original integrals. Differentiating numerator and denominator gives eq. 5.i by the quotient rule. Thus the bound for the renewed bond pointing must include the direct insertion derivative and the connected term with the potential derivative on its actual read. The fixed readout and vacuum evaluation are bounded linear operations in the pointed member norm, so differentiation commutes with them. Differentiating the two moving operations by linearity and the product rule gives eq. 5.ii. In the frozen source presentation both additional reference derivatives vanish. □

Write X^+ for the complete enlarged source read of a domain. The geometric input is the native one-layer enlargement with one fixed additional bare-plaquette reach for barycentres. All local operations are evaluated on complete determining sets. The actual filing rule puts a fine vertex in a regular output domain only when the coarse hull of its complete one-layer read lies in that domain. Boundary and retained terms use their own full collars. At the filing of an output (X, z) the enlarged reads of its vertices, including earlier complete reads, lie in X^+. The same condition holds for the read of an inherited boundary summand. A retained component uses its own collared carrier and its fixed stored variables. When it is included in a new filed output, that full read is part of the output carrier. These are the geometric admission conditions on the formation. They also give

$$\sup_{p\in X^+} d(x(p), x(p_z)) \le CMh_j(1 + d_j(X)). \tag{5.iii}$$

The path of a cube tree has length at most its number of edges times Mh_j, and the endpoint and collar errors are at most a fixed multiple of Mh_j.

Lemma 5.4 (Comparison on a local data line)**.** Suppose a filed output reads only its complete input stock on X^+. Compare a variable source W there with a constant v, and put

$$m = \max\{ \sup_{p\in X^+} |W(p) - v|,\ c_M h_j \operatorname{Lip} W\}.$$

On the independent local data box suppose the line from the constant data to the sourced data, with the common reference fixed, is admitted for $|t|m < r$. When that disc contains 1, the line has the actual sourced output at $t = 1$. Suppose the normalizer-subtracted output is holomorphic on that disc and is bounded on the line and at both actual endpoints by $Be^{-\gamma d_j(X)}$. Its endpoint difference is at most

$$2B \min\{1, m/r\} e^{-\gamma d_j(X)}. \tag{5.iv}$$

With $v = W(p_z)$, $\gamma \geq \kappa + 1$, and $j \leq n$, the right side is bounded by

$$CBr^{-1} L^{-(n-j)} \|w\|_{\flat,K-n}\, e^{-\kappa d_j(X)}. \tag{5.v}$$

■

Proof. For $0 < m < r$ subtract the value at $t = 0$ and apply Schwarz on $|t| < r/m$. For $m \geq r$ use the two endpoint bounds. At $m = 0$ the local input reads coincide. With the anchor value as constant, eq. 5.iii bounds m by $C(1 + d_j(X))c_M h_j \operatorname{Lip} W$. Absorb the diameter using the spare unit of decay and use $h_j/h_n = L^{-(n-j)}$ and the native source norm at depth $K - n$. This proves eq. 5.v. The line is in the independent data box. Interpolation in the physical source class alone would not give this estimate up to its boundary. □

Proposition 5.5 (Formation and old-origin summation)**.** Use the actual regular formation, component operations and restrictions just described. At level n assume the ordinary window estimates on their admitted field domains. Let $\rho_{jn} \geq 1$ bound the ratio of the radius required by the output window to the available radius of its old member; the value one is used when that ratio is smaller. A single normalized regular origin at level j, with its own vacuum and action subtraction, contributes at most $CG_*\rho_{jn}^3 L^{-(4+\alpha_H)(n-j)}$ to the window, where $\alpha_H > 0$. A vacuum-subtracted defect or remainder with norm B_j contributes at most $CB_j\rho_{jn}^2 L^{-4(n-j)}$. The bounds include the sums over domains through that origin and $\rho_{jn}^2 \leq C(1 + n - j)$. Assume the complete filed data boxes admit their constant and variable endpoints, and that their local lines satisfy the hypotheses of lemma 5.4, including holomorphy on the full independent disc. Assume the actual normalizer-subtracted pointed output is bounded by $\frac{1}{2}e^{-(\kappa+1)d_j(X)}$ on those lines and endpoints, and remainders and boundary outputs have their respective ordinary rooted bounds there. On each source family the successive outputs are holomorphic, are real for real sources and fields, and depend on the source only through their complete bare plaquette reads. The difference from the selected constant-source member is a vacuum-subtractable defect of norm at most $Cr^{-1}\|w\|_{\flat,K-n}L^{-(n-j)}$ at age $n - j$. If the vacuum-subtracted remainder norms are at most $C_R\theta_j^{-p}$, $p > 2$, and the lower drift in proposition 4.1 holds, the total ordinary input from these three families is bounded by

$$CG_* + Cr^{-1}\|w\|_{\flat,K-n} + CC_R\theta_n^{2-p}. \tag{5.vi}$$

The bounds hold for the sourced filing with a different selected member at each base point. ■

Proof. At the bare stage the source enters only as the values of W at plaquette points. At a subsequent stage first form the actual regular output with the ordinary rule. The complete input reads of a connected term are inside its filing stock. The source values of an earlier read are there too by the geometric inclusions. Products, fixed coordinate substitutions, integration and normally convergent rooted sums therefore give holomorphy and dependence on these values alone. Conjugation commutes with these operations and fixes their real reference branches. The same reasoning on each complete retained carrier gives the boundary and remainder assertion, using the supplied domination for those operations. Carried terms are restricted with their original carriers

and branches. Thus equality of two source representations on the complete read gives equality of the output.

At each new pointing select the whole-lattice member at its anchor value. It is formed with the same ordinary interior rule. When W is constant on the complete read there is equality with this member on the common field window. Otherwise lemma 5.4 gives its defect bound. Taking out the member's own coefficient and reinstating it by eq. 4.iii gives the exact exponent, by the pointwise identity for the weights. Thus at the next step the actual sourced input consists of normalized selected members, defects and remainders on their declared domains. Bounds on these domains and on their local lines are needed for the construction. Neither holomorphy nor summation by itself supplies those admissions.

A level-n window contains at most $CL^{4(n-j)}$ origins of level j. In each of the window estimates this cancels the volume factor. With $a = n-j$, the regular sum is bounded by $CG_* \sum_{a\geq 0}(1+a)^{3/2}L^{-\alpha_H a}$. The defect sum is bounded by $Cr^{-1}\|w\|_{\flat,K-n}\sum_{a\geq 0}(1+a)L^{-a}$. For remainders, $\theta_j \geq \theta_n + \lambda a - C_0$ and a large fixed floor bound the sum by a constant times

$$C_R \sum_{a\geq 0}(1+a)(\theta_n/2 + \lambda a)^{-p} \leq CC_R\theta_n^{2-p}.$$

The last inequality follows by comparison with the integral of $(\theta_n/2 + \lambda t)^{1-p}$, which is finite for $p > 2$. Every constant used the supremum of the member norms and is independent of the sitewise selections. This proves eq. 5.vi. □

Admission of the comparison lines. For full ordinary source radius $2r$, use the constant companions on $|v| < 4r$. The argument of proposition 4.1 on this larger disc uses $6r$ of half-product room, with the marked estimate on the corresponding actual companion disc. It gives the same summable bound on the radial slopes. Cauchy therefore gives a Lipschitz constant $2d_i$ for b_i on $|v| < 2r$. This reserve is chosen in the supplied local formation before the inverse-coupling floor.

Here is a sufficient check of the independent line hypotheses in proposition 5.5. Suppose the analytic ordinary rule admits site-source and running-weight slots of radius $4r$. On the native carriers let B_i bound the normalizer-subtracted regular, boundary and remainder slots, with their actual rates and remainder small factors. Assume the enlarged box admits, about each constant-source centre, perturbations of norm $2B_i$ in these slots, measured with their native rates and small factors. In a box about zero, $3B_i$ of absolute room suffices. Give the independently selected members the same comparison reserve. The ordinary formation estimates on this box must bound the output regular legs in the following exponential norms. With the decay rate $\kappa+1$, the interaction leg costs Cs_k times the sum of the old-origin input norms and $Cs_k\alpha_k^2$ times the Wilson shift norm. The intrinsic, coordinate and cutoff legs together cost $CM^4\theta_k^{-1/2}\mathfrak{b}_k^3 + C\theta_k^{-1/4}$, where $\mathfrak{b}_k$ is at most a fixed power of $\log\theta_k$. The free quadratic correction costs $C\mathfrak{t}_k/\theta_k$ on the same rate, where $\mathfrak{t}_k$ is the real tube radius. These are estimates for the actual formed legs on their local cores. The two reference Gaussian contributions being subtracted are common to the line. For a retained component use the absolute excess recursion and domination on this enlarged box, as in the retained-integral hypotheses below.

Fix a complete output stock and a constant comparison value v. On the stock interpolate the site values, selected regular members and their own coefficients affinely from constant to variable data. Retain the fixed reference entries elsewhere. Take m as in lemma 5.4 at the output scale. For $|t|m < r$ a site value has modulus at most $3r$. The sum of the coefficient departures from the zero-source values, including the interpolated differences, is at most $(2r + 2r)\sum_i d_i \leq 2r/3$. Thus the running weight shift fits the $4r$ slot. By induction the input regular and boundary differences at their respective complete reads are bounded by $2B_i m/r$. Multiplication by t fits their reserved room. The same holds for the remainder slots with their own small factors. An old defect vanishes at the constant endpoint. Its older bound from eq. 5.v, applied before the final source norm comparison,

is $Cr^{-1}h_i \operatorname{Lip} W$. Thus its line norm at output level $k+1$ is at most $CL^{-(k+1-i)}$. These checks use only complete reads inside the chosen stock. The linear restrictions of a boundary sum are applied to its individual summands. A retained excess is generated by its actual operation on the inherited lines, including the reference contribution and all earlier excess terms. This uses domination on the full input line. Interpolating only a part of an exponent could lose a cancellation between terms with large constant parts.

The ordinary origin sums are now uniformly bounded by the geometric, derivative-of-geometric and power sums in eq. 5.vi. The displayed leg estimates give a normalizer-subtracted regular output on the line bounded by

$$C(s_k + M^4\theta_k^{-1/2}\mathfrak{b}_k^3 + \theta_k^{-1/4} + \mathfrak{t}_k/\theta_k)e^{-(\kappa+1)d_{k+1}(X)}.$$

A sufficiently small relative tube and sufficiently large floor make the prefactor at most $1/2$. The ordinary bounded output estimates on the enlarged box give the other B_{k+1} bounds. Applying lemma 5.4 recovers the required slopes for the next stage. At the first level the bare site interpolation is affine and there are no old regular or retained entries. This inductive check uses the local estimates on the actual enlarged box. It deduces source-line bounds from them rather than extending the physical-source disc by assertion.

5.1. **First source derivatives and the near-window domains.** The near-window conclusion below uses a complete nonwrapping background-plus-fluctuation datum of size at most $C\delta_n$ at level n, including its gauge and remote reads. We specify the local background input for an origin of level $j \le n$. Put $\sigma = L^{-(n-j)}$ and $D = 1 + d_j(X)$. At each fixed pointing z use one affine curvature $F_{j,z}$ and its linear affine pair L_F in the common old window chart. Every domain in the pointed total, including the tent action, uses the restriction of this same pair. In the old pair norm the evaluated near pair P satisfies

$$\begin{aligned} \|L_F\|_X &\le CD^b\delta_n\sigma^2, \\ \|P - L_F\|_X &\le CD^b(\delta_n\sigma^{2+\alpha_H} + r_{B,j}^{-1}\delta_n^2\sigma^4), \qquad \alpha_j/r_{B,j} \le C, \end{aligned} \tag{5.vii}$$

where $\alpha_H > 0$ and $r_{B,j}$ is the old independent datum radius. The settled lower plaquette bound is $C\delta_n\sigma^2$. The common upper and remote reads have amplitude $C\delta_n$. Use the actual weighted response on those reads and the full absolute datum ceiling, with every complex substitution admitted. The affine norm bound in eq. 5.vii holds also on domains omitted from the near chart. Any such omission used in the affine quadratic cancellation has length at least $c/\sigma - C$.

Each local old regular and defect member, at its individual (X, z), is invariant under simultaneous constant adjoint action on its admitted pair chart. For the complete constant-source member use the affine coefficient identity at each pointing of [3, Lemma 7.1], including its scalar lattice covariances. The identity for complex affine curvatures follows from its real version by polynomial complexification of the quadratic forms. Separately from the datum radius, the old regular and defect members, and their first-source jets used below, are holomorphic on pair charts containing a ball of radius $c\alpha_j$ about the trivial pair in their declared norm. Their norm bounds hold on these balls and on all actual evaluated pairs, even when an evaluated pair lies outside the smaller Taylor ball. The pointed action is holomorphic on the field projections of these balls and is admitted on all actual evaluated fields. Its norm is $O(\alpha_j^2)$ on both sets on its fixed bounded carrier. The affine readout of a unit curvature has pair norm bounded by $C(1 + d_j(X))^b$ on each old domain. Its transported Hessian uses the unit coefficient convention of eq. 4.i. The field estimates for the complete window and its independent current are the local inputs here. The size of a fluctuation alone does not supply them.

Proposition 5.6 (First-source response at fixed reference)**.** Assume the lower drift in proposition 4.1 and $\theta_i - \theta_j \le C(1+j-i)$ for $i \le j$. For a source direction f, write $D_j[f] = \partial_t E_j^{tf}|_{t=0}$ for the pointed regular jet, and let A_j be its operator norm from $\|f\|_{\sharp,K-j}$ to $\|\cdot\|_{j,\kappa+1}$. The initial regular list is empty. Use the actual one-step vertex and renewed pointing estimates on complete admitted reads, with the following weights. The Wilson direction costs $C\alpha_k^2 s_k$ times the shift norm. The fluctuation difference of an old normalized regular jet has weight $Cs_k\omega_{k+1-j}$, where $\omega_a \ge 0$ and $\sum_{a\ge1}\omega_a < \infty$. An old vacuum-subtracted defect or admitted aged remainder has weight $Cs_k(\alpha_k/\alpha_j)^2$. For every remainder of origin j and every source direction f, its actual reads are admitted on the line $|t|\|f\|_{\sharp,K-j} < r$, its value is holomorphic there, and its norm there is at most $C_R\theta_j^{-p}$ with $p > 3$. This radius uses the physical scale of that origin, not the unit-scale test norm. For a sufficiently large floor these estimates imply

$$A_j \le C\alpha_{j-1}^2 s_{j-1} \qquad (j \ge 1). \tag{5.viii}$$

If the near-window inputs of section 5.1 hold, the total of the renormalized near regular and defect jets per level-n window is at most $C\delta_n^2 s_n \|f\|_{\sharp,K-n}$, where $\delta_n^2 s_n = \theta_n^{-1}(\log\theta_n)^{3p_0-q}$. Additional winding, marked and field-independent remote tails retain their own bounds. ∎

Proof. Subtract $f(p_z)D_j[1]$ from the jet at (X,z). By eq. 5.iii, the profile $f - f(p_z)$ on the complete read has size at most $Ch_j(1+d_j(X))\operatorname{Lip} f$. Clip its real and imaginary parts at that bound outside the read. The clipped profile has unchanged values on the read and Lipschitz constant at most a fixed multiple of $\operatorname{Lip} f$. The jet is linear and local. The clipped sharp norm and the spare decay unit give the bound $CA_jh_j\operatorname{Lip} f$ for the defect at rate κ. At level $k+1$ this is at most $CA_jL^{-(k+1-j)}\|f\|_{\sharp,K-k-1}$. The derivative of the selected member's coefficient is its own $b_j'(0)f(p_z)$ by linearity of the fixed physical readout. Differentiating eq. 4.iii bounds the actual Wilson shift derivative by $(1+\sum_j d_j)\|f\|_\infty$. Cauchy on the remainder disc bounds the remainder derivative by $Cr^{-1}\theta_j^{-p}\|f\|_{\sharp,K-j}$.

Let $r_{kj} = \alpha_k/\alpha_j$. The one-step bounds give

$$A_{k+1} \le Cs_k\left[\alpha_k^2 + \sum_{j=1}^{k}\omega_{k+1-j}A_j + \sum_{j=1}^{k} r_{kj}^2 A_j L^{-(k+1-j)} + r^{-1}\sum_{j=0}^{k} r_{kj}^2\theta_j^{-p}\right]. \tag{5.ix}$$

The norm of f at an earlier level is bounded by its norm at level $k+1$. The adjacent weights have bounded ratios. For large θ, both $\alpha(\theta)^2$ and $s(\theta)$ decrease, and $\theta_j \ge \theta_k - C_0$ for $j \le k$. Thus $\alpha_{j-1}^2 s_{j-1} \le C\alpha_k^2 s_k$. With r_{kj}^2 inserted the same bound follows from $r_{kj}^2\alpha_{j-1}^2 s_{j-1} = \alpha_k^2(\alpha_{j-1}/\alpha_j)^2 s_{j-1}$. The upper drift gives $r_{kj}^2 \le C(1+k-j)$. The last sum in eq. 5.ix is $O(\theta_k^{2-p})$ as in eq. 5.vi. Since $p > 3$ it is $O(\alpha_k^2)$ at fixed r. Under the inductive bound $A_j \le C_*\alpha_{j-1}^2 s_{j-1}$ the right side is at most $\alpha_k^2 s_k(C_1 + C_2C_*s_k)$. Take $C_* \ge 2C_1$ and make $C_2 s_k \le 1/2$ by the floor. Empty sums start the induction.

For the near conclusion, the central axial comb uses at most $C\sigma^{-1}$ plaquette factors of size $C\delta_n\sigma^2$. Their product has link logarithm of size $C\delta_n\sigma$. The cutoff extension costs $C\delta_n$ in the full row norm. The remote rows have the proportional amplitude $C\delta_n$, so their weighted first response is bounded by $C\delta_n e^{-c/\sigma}$ and fits the first error in eq. 5.vii. The central half-disc condition is uniform in age because

$$\delta_n\sigma/r_{B,j} \le Cs_n(1+n-j)^{1/2}L^{-(n-j)}.$$

The right side is small at the chosen floor. The remaining absolute core inclusions are those of the standing window data.

The regular member and the defect both have zero linear term at the trivial pair. The ordinary scalar source preserves their constant adjoint invariance at each pointing, and the adjoint representation has no invariant linear functional. In the complete regular pointed sum the Hessian on L_F is cancelled by its own tent action at that pointing. The coefficient of a member of norm G is

bounded by $CG\alpha_j^{-2}$. To see this, apply its Hessian to the unit affine curvature used for the pointed coefficient. The affine pair on a domain grows at most polynomially in its length, the pair ball gives the factor α_j^{-2} by Cauchy, and the domain sum converges by the strict entropy margin. The unit action Hessian is fixed. Thus the own tent subtraction has norm at most CG, by its $O(\alpha_j^2)$ bound. This estimate uses the fixed affine readout on the complete pointed member.

Put $\rho_j = \delta_n/\alpha_j$ and $u = CD^b\rho_j\sigma^2$. For small u, eq. 5.vii places P in the half ball of the old pair chart. For a local term of norm g_X including its exponential decay, Cauchy's estimate gives a Taylor remainder Cg_Xu^3. The difference between its quadratic values on P and L_F is at most

$$Cg_XD^{b'}(\rho_j^2\sigma^{4+\alpha_H} + \rho_j^3\sigma^6).$$

The two products of the affine field with the errors give these terms directly. The squared errors satisfy the same bound, using u small and $\sigma \le 1$. For u above a fixed positive threshold, use the actual core bound and the affine quadratic bound Cg_Xu^2. Their sum is bounded by Cg_Xu^3. For the vacuum-subtracted defect, the zero linear term and the same pair bound give Cg_Xu^2 for small u, and the actual core bound gives it for large u.

Apply the affine cancellation at each pointing before multiplying by $f(p_z)$. If the near part omits domains, the missing affine Hessians are bounded by the same $Cg_XD^{b'}\rho_j^2\sigma^4$. Their lengths are at least $c/\sigma - C$, so the rooted sum adds $O(e^{-c'/\sigma})$ and fits the $\sigma^{4+\alpha_H}$ term. This use of the omitted affine values imposes no estimate on an actual far evaluated field. All polynomial factors in D are summable at the reserved decay. There are $O(\sigma^{-4})$ old base points in the window. After this count the regular factors are $\rho_j^2\sigma^{\alpha_H}$ and $\rho_j^3\sigma^2$, and the defect factor is ρ_j^2.

The constant-source direction at a base point is $f(p_z)D_j[1]$, of norm at most $A_j\|f\|_\infty$. By the proved amplitude, $A_j/\alpha_j^2 \le Cs_n$ and $\rho_j \le Cs_n(1+n-j)^{1/2}$. With $a = n-j$, the sum of the regular terms is bounded by

$$C\delta_n^2 s_n\|f\|_\infty \sum_{a\ge 0}\big(L^{-\alpha_H a} + s_n(1+a)^{1/2}L^{-2a}\big).$$

The defect sum is at most $C\delta_n^2 s_n \operatorname{Lip} f \sum_{j\le n} h_j$, which is $C\delta_n^2 s_n\|f\|_{\sharp,K-n}$ by the mesh ratio. The two estimates prove the stated near bound. □

5.2. **Sources in retained operations.** The complex source changes the integrand of a retained operation on its fixed stored variables. Write $\Phi_\omega(\xi;P)$ for the reference exponent of a record ω at zero source, and write Ψ_ω for its source excess. The recorded reference measure μ_ω, the integration set and the real stored-field cut are fixed. The exterior argument P ranges over the actual admitted set over ξ. This set includes the prescribed test compatibility for a drop-out. The Wilson contribution carried by another factor stays on that factor.

Lemma 5.7 (Source excess and retained mass)**.** Consider a finite nested family of actual retained operations on a common admitted source or local-data disc. For each record ω let n_ω^0 count fresh charged sites in their own scales, and let $n_\omega = n_\omega^0 + \sum_{\omega'} n_{\omega'}$, summing over the swallowed records. Assume the actual source-sensitive core Wilson term has the form $A(\widehat{c}\,\upsilon 1_X, U'')$, with $|\widehat{c}| \le \bar{c}$, $0 \le \upsilon \le 1$, and real stored background U''. On each nonbare layer j where the cut is nonzero, assume its plaquette defects are bounded by $C\eta_j^2$, where η_j is the bare mesh in those layer units, with at most $C\eta_j^{-4}$ such plaquettes per fresh site. On the bare layer use the uniform plaquette bound and bounded incidence. Omit this core term when the operation has none.

Suppose the actual fresh collar terms $V_X^{\mathrm{col},W}$ and $V_X^{\mathrm{col},0}$ have the sum of absolute bounds at most $C(1+\bar{c})n_\omega^0$. This includes their constant parts. The kept responses of old carrier functions use one common map. On a kept carrier R, applying the response to its excess function ψ_R gives its value at the actual stored pair $q(\xi)|_R$ plus a correction $\delta_R[\psi_R]$. Assume the absolute sum of these corrections is at most $C(1+\bar{c})n_\omega^0$ on the complete reads. Each stored pair lies in the admitted domain of the corresponding old exponent. For each record use a fixed Borel integration domain with a positive sigma-finite reference measure. The evaluated integrands are jointly measurable on products with compact parameter sets. The evaluated exponents and reference factors, including the re-evaluations of old exponents, are holomorphic in the given parameters for almost every stored tuple on that domain.

Then the source excess of the complete record satisfies

$$|\Psi_\omega| \le C(1+\bar{c})n_\omega. \tag{5.x}$$

The constant is independent of nesting depth. Suppose the fixed reserve weight satisfies $v(\omega) \ge c_v n_\omega$ for a fixed $c_v > 0$. Let $\varpi(\omega) \ge 0$ be the actual record weights, and suppose their real weighted mass satisfies

$$\sum_\omega \varpi(\omega) \int^* e^{cv(\omega)} \sup_P |e^{\Phi_\omega(\xi;P)}|\, d\mu_\omega(\xi) \le M_c \tag{5.xi}$$

for a fixed c large enough to pay eq. 5.x. Then the source-dependent operation tested against a fixed bounded Borel function of the stored variables has absolute value at most $M_c\|F\|_\infty$. It is holomorphic on the same admitted parameter disc for a fixed bounded F. Any further visible-record price uses the same restricted mass at a correspondingly larger reserve. ∎

Proof. Subtract the complete zero-source exponent on the same stored tuple. The fixed reference quadratic cancels. In each kept response put its value at $q(\xi)|_R$ back in the corresponding old exponent. The corrections remain in the fresh term, so the excess is

$$\Psi_\omega = \Psi_\omega^{\mathrm{new}} + \sum_{\omega'} \Psi_{\omega'}(\xi'; q(\xi)),$$
$$\Psi_\omega^{\mathrm{new}} = A(\widehat{c}\,\upsilon 1_X, U'') + V_X^{\mathrm{col},W} - V_X^{\mathrm{col},0} + \sum_R \delta_R[\psi_R].$$

This decomposition uses the complete older exponent at its read. It does not combine independently chosen pieces of old exponents. On a nonbare layer the identity $a_p(U'') = \frac{1}{2}\|U_p'' - I\|_{\mathrm{op}}^2$ gives a bound $C\eta_j^4$ per plaquette. Multiplying by the incidence bound cancels the mesh factor and costs at most $C\bar{c}$ per fresh site. The bare layer gives the same site bound using $a_p \le 2$. Add the two supplied fresh bounds to obtain $|\Psi_\omega^{\mathrm{new}}| \le C(1+\bar{c})n_\omega^0$. At a leaf this already proves eq. 5.x. For other records add the older estimates at their own counts. By additivity the new constant is the maximum of the fresh and old constants. This proves the bound by induction.

For each ξ and each admitted exterior argument, the modulus of the source multiplier is at most $e^{cv(\omega)}$. Use this bound before taking the exterior supremum. Equation eq. 5.xi then gives an integrable majorant, including the nonnegative record weights ϖ. Integrating after multiplication by F gives the stated bound. The same majorant on compact subdiscs permits integration of the holomorphic integrand around a triangle. Fubini gives zero, so Morera's theorem gives holomorphy

in each parameter. Local boundedness then gives joint holomorphy on a finite parameter product. The upper integral is used only to bound the supremum in eq. 5.xi. Choose measurable majorants with total integral arbitrarily close to that bound. The evaluated integrands themselves are measurable by hypothesis. Restricting records first and raising c pays a further bounded price per recorded site with the same argument. □

The absolute collar bounds in this lemma cannot be replaced by oscillation bounds in the exterior pair. Oscillation does control the kept-response corrections after their common value at the stored pair has been subtracted. If the actual normalized carrier oscillation at the next operation is at most $b_* + \epsilon s$ when the inherited bound is s, choose $c_* \geq 2b_*$ and keep the formation domain $s \leq 4c_*$. For $8\epsilon \leq 1$ the new bound is at most c_*. In the ordinary local estimate the coefficient ϵ decreases as a positive inverse power of $\log \theta_j$ at fixed other constants. This is a separate closure estimate on the common enlarged data domain. It gives no bound on an omitted field-independent collar value.

5.3. **A common majorant and the nonvanishing boundary.** The retained mass hypothesis uses a common reference coordinate system before taking a supremum over exterior data. The following calculation supplies one sufficient envelope when such coordinates are available. It does not construct those coordinates.

Lemma 5.8 (A common Gaussian majorant)**.** Let $b \in \mathbb{R}^m$, where m is a nonnegative integer, $N \geq 0$ and $m \leq c_0 N$. Let σ index real symmetric matrices $Q_\sigma \geq \gamma I$, where $\gamma > 0$. Suppose $c_0, C_0 > 0$, $|c_\sigma| \leq C_0^m$, S_σ is Borel, and the functions V_σ and $H_\sigma(z, \cdot)$ are Borel. For fixed real constants B_0, δ, assume

$$\operatorname{Re} V_\sigma(b) \leq B_0 N + \gamma |b|^2/4, \qquad \operatorname{Re} H_\sigma(z, b) \leq \delta$$

on the parameter set being considered. Then the densities $1_{S_\sigma} c_\sigma e^{-b^{\mathsf{T}} Q_\sigma b/2 + V_\sigma(b) + H_\sigma(z,b)}$ have the common Borel absolute majorant

$$C_0^m e^{B_0 N + \delta} e^{-\gamma |b|^2/4}.$$

Its integral is at most $e^{CN+\delta}$, with C depending only on c_0, C_0, B_0, γ. The same bound controls the upper integral of the supremum of the absolute densities. ■

Proof. The lower quadratic bound leaves $-\gamma|b|^2/4$ after paying the quadratic part of the bound on V_σ. Taking absolute values gives the displayed majorant pointwise. Its integral is

$$C_0^m e^{B_0 N + \delta} (4\pi/\gamma)^{m/2}.$$

The bound on m absorbs its two dimension factors in e^{CN}. The majorant is Borel even if an uncountable supremum of the densities is not. □

Parameter-dependent changes of variables with separate bounds do not supply this common envelope. Nor does holomorphic dependence on a background parameter by itself ensure a nonzero integral.

Lemma 5.9 (A phase can cancel a tested integral)**.** Let w_0 be integrable on a measure space, with $W_0 = \int w_0 \, d\mu \neq 0$. Let $x \geq 0$ be measurable, and suppose the pushforward of $|w_0| \, d\mu$ under x is atomless. There are $c > 0$ and $\lambda \in [-\pi, \pi]$ for which

$$\int w_0 e^{i\lambda 1_{\{x > c\}}} \, d\mu = 0.$$

■

Proof. Put $A(c) = \int_{x \leq c} w_0 \, d\mu$ and $B(c) = W_0 - A(c)$. Atomlessness and integrability make these functions continuous, with $A(0) = 0$ and $A(c) \to W_0$. Thus $|A(c)| - |B(c)|$ changes from a negative to a positive value at some finite $c > 0$. At a zero, $|A(c)| = |B(c)| > 0$; otherwise $W_0 = 0$. Choose $e^{i\lambda} = -A(c)/B(c)$ and split the integral at c. □

The phase in this example can be independent of every background parameter, hence holomorphic in those parameters. If x is gauge invariant, the phase is gauge invariant too. Bounds on admitted switch and displacement domains, or the explicit local nonvanishing estimates used below, supply additional information that this example lacks. A logarithm of a retained or full partition integral must use such information.

6. Rooted counting for source reads

At spacing $a_s = L^{-s}$ the localization cubes have side Ma_s, with the fixed M of section 2. Take tori of side $\ell = 2L^m$ with $m \geq m_0$, so the localization grids at every retained level tile the torus. A nonempty domain X is indexed by its set of cubes. Let $m_s(X)$ be the minimum number of grid edges in a tree meeting every cube of X, allowing a one-vertex tree of length zero. A full source read X^+ lies within a fixed number of layers of these cubes. The enlargement includes any reads of plaquette centres.

Lemma 6.1 (Rooted domain counts)**.** For a cube Q and an integer $b \geq 0$, the number of domains X containing Q with $m_s(X) = b$ is at most $16\, 8^{2b} 2^{16(b+1)}$. In particular, if κ exceeds the rooted entropy exponent κ_D, then for some $C, c > 0$,

$$\begin{aligned} \sum_{\substack{X^+ \cap B_u \neq \varnothing \\ m_s(X) \geq t}} e^{-\kappa m_s(X)} &\leq C(1 + u/a_s)^4 e^{-ct}, \\ \sum_{X} e^{-\kappa m_s(X)} &\leq C\ell^4 a_s^{-4}, \qquad \operatorname{diam} X^+ \leq C a_s (1 + m_s(X)). \end{aligned} \tag{6.i}$$

Here B_u is any ball of radius u, $t \geq 0$, and the distances are those of the periodic quotient. The same estimates hold for indexed families with a total rooted count $Ce^{\kappa_D b}$, including their multiplicity. ■

Proof. A grid edge meeting a closed cube has an endpoint in the cube. Thus a minimizing tree with b edges has a vertex in every cube of X. Start at a corner of Q in the tree, and walk over each tree edge twice. There are at most 16 starting corners, 8^{2b} possible walks, and $16(b+1)$ cubes incident to their vertices. The choice of a subset specifies X. On a quotient there can be fewer distinct cubes and there are still at most eight oriented steps at each vertex. The same count applies.

If X^+ meets the ball, some cube of X is within CMa_s of it. The number of possible such root cubes is at most $C(1+u/a_s)^4$. At each root the sum over lengths at least t is geometric, by $\kappa > \kappa_D$. Summing instead over all root cubes gives the global count. Two points in X^+ are within CMa_s of vertices of a minimizing tree, whose distance along the tree is at most $Ma_s m_s(X)$. This proves the diameter bound. □

7. Local source energies and the common normalization

For each level j and localization domain X, write $E^{(j),W}(X, P, z)$ for the newly formed regular member at its pointing z, before its vacuum and classical-action subtraction. Write $R^{(j),W}(X, P)$ for the completed new remainder before its vacuum subtraction. Use Δ_W for the difference from the same member at zero source. Define

$$e_{K-j,X}(W) = \sum_{z \in X \cap \Lambda_j} \Delta_W E^{(j)}(X, P_\circ, z) + \Delta_W R^{(j)}(X, P_\circ), \qquad j \geq 1. \tag{7.i}$$

Here Λ_j is the lattice at level j. At $j = 0$ put $e_{K,X} = 0$. In the source normalizer take the point term $U(W) = 0$.

Proposition 7.1 (Locality, entropy and the vacuum difference)**.** Use the sourced formation of proposition 5.5 on the ordinary ball of a prefix, with its complete reads. Assume that the regular outputs at $P_\circ$, after subtraction of their common frozen-reference term $N^{(j)}$, have uniform bound $Ce^{-(\kappa+1)d_j(X)}$. Here $N^{(j)}$ is the actual regular normalizer and assigned fixed-real-weight ratio term, identical as a function for source W and source zero. Assume the corresponding remainders have bound $C\theta_j^{-p}e^{-\kappa d_j(X)}$ with $p>0$. Index a domain by its union of localization cubes, once per union at that level. Let κ exceed the rooted entropy rate for this indexing. Then the local energies in eq. 7.i are holomorphic, real for real sources and zero at zero source. They depend on the bare plaquette values in X^+ alone and satisfy

$$|e_{K-j,X}(W)| \le Cr^{-1}\|w\|_{\flat,K-n}\, e^{-\kappa d_j(X)} \qquad (1\le j\le n) \tag{7.ii}$$

on the prefix stopped at n. The bound holds also with $m(X)$, the shortest cube-edge tree length allowed to leave X, in place of $d_j(X)$.

Use the once-counted vacuum convention of [4]. For this prefix let $V=E_{(n)}$ be the unsourced vacuum and $\mathcal{E}(W)=\sum_{j\le n,X} e_{K-j,X}(W)$. The all-small-field scalar is $V+\mathcal{E}(W)$. In any history whose scalar shares are obtained by the actual filing, let $F_h(0)$ be its unsourced scalar exponent and let $\mathcal{E}_h(W)$ be its retained source share. With this normalization the scalar exponent is

$$F_h(0)-V-\{\mathcal{E}(W)-\mathcal{E}_h(W)\}. \tag{7.iii}$$

Suppose each missing pointing and each missing remainder at level j is incident by its own read to the bounded-layer neighbourhood of a paid cube of that label. The source difference in braces costs at most $C\sum_{j\le n}\#Z_j^+$ on the ordinary ball, where Z_j^+ denotes the paid cube set with its fixed collar. ■

Proof. In the once-counted scalar the Gaussian, gauge and Haar factors of the frozen reference do not vary with the source. All Wilson terms vanish at the trivial pair. Each regular trivial value is assigned at its pointing, and the new remainder value is counted once after closure. Taking their differences gives eq. 7.i and the asserted all-small-field scalar. The same subtraction in an individual history gives its own source share. Subtracting the all-small-field scalar then gives eq. 7.iii with the displayed sign.

Local formation gives holomorphy, reality and the specified reads. The term $N^{(j)}$ stays fixed when the source changes. At $P_\circ$ this gives

$$\Delta_W E^{(j)} = \Delta_W(E^{(j)}-N^{(j)}).$$

The remaining trivial value is retained in the vacuum readout. For the bound, use the holomorphic source line tW up to the ordinary source radius. Subtract the value at $t=0$ and apply Schwarz to the uniform bound. Each regular difference then costs at most $Cr^{-1}\|w\|_{\flat,K-n}e^{-(\kappa+1)d_j(X)}$, and the remainder difference costs at most this source factor times $C\theta_j^{-p}e^{-\kappa d_j(X)}$. The zero source gives zero directly. A tree with $d_j(X)$ edges meets at most $C(1+d_j(X))$ cubes and therefore at most $CM^4(1+d_j(X))$ pointings. The spare unit of rate absorbs this factor. Since $m(X)\le d_j(X)$, eq. 7.ii also holds at the shorter length.

For the last assertion, assign each missing term to one of its incident paid cubes. For one paid cube its possible roots lie in a fixed collar, and the domain sum through any such root is bounded by the strict entropy gap. Counting possible pointings uses the spare regular rate just as above. A remainder has one term per domain. The uniform bound per paid cube gives the cost stated in the proposition. □

The cube at depth s has side Ma_s, where $M=L^{m_0}$ and $a_s=L^{-s}$. On $\ell=2L^m$, $m\ge m_0$, the number of these cubes in each direction is $2L^{m-m_0+s}$. This gives even counts on the whole cofinal sequence. Every fixed board with side a power of L and integral in bare lattice units fits in all sufficiently large members of the sequence. Domains at each of the actual scales $0\le s\le K$ have the

indexing used in the rooted counts. The one-layer enlargement, including the fixed bare-barycentre reach, stays within a fixed number of neighboring cubes since $a \le a_s$. Thus the support-range and multiplicity conditions in the subsequent geometric estimates refer to the actual energy sum. If a filing uses additional domain types, their multiplicity must be paid in the same entropy estimate.

The physical normalized integral at a stop is $\widetilde{Z}(W) = e^{-V-\mathcal{E}(W)}Z(W)$. This uses the exact history mass in eq. 3.i. On two admitted common prefixes the same local formation pulls back the same bare plaquette arrays. The equality of the corresponding source energies and scalar shares on the common reads follows from their definitions. Under an admitted change of lattice units this statement uses the relabeling of the frozen tests, measures, carriers and pointing data of [4]. At a stop, the natural source norm is $\|w\|_{\flat,K-n}$. With unchanged physical units the site count is $\ell^4 h_n^{-4}$, and with the units of the stop it is the fourth power of the dilated side. These counts agree. The upper and lower bounds on this normalized integral require the retained-integral estimates and the unsourced floor.

If this same real run is admitted through a later level K, write $\mathcal{E}_{(n)}(W) = \mathcal{E}(W)$ for the source sum through its stop n. The two vacuum conventions are related by the exact identity

$$e^{-E_{(K)}-\mathcal{E}_{(n)}(W)}Z(W) = e^{E_{(n)}-E_{(K)}}e^{-E_{(n)}-\mathcal{E}_{(n)}(W)}Z(W). \tag{7.iv}$$

Assume the full-level scalar convention and the bounded remaining scalar prices of [4, Proposition 2.5], with $\theta_j \ge 1$ and $\theta_i - \theta_j \ge -C_{\mathrm{exc}}$ for $i \le j$. That proposition bounds the positive multiplier by

$$e^{-CN_n} \le e^{E_{(n)}-E_{(K)}} \le e^{C(1+\log\theta_n)N_n}, \qquad N_n = \ell^4 h_n^{-4}.$$

Thus both normalizations give an upper bound and a zero-source floor with the corresponding adjusted stop-scale exponents. The source sum still ends at n in both conventions.

7.1. **The neighborhood where a partition logarithm exists.** An upper bound for a holomorphic partition function and a positive value at zero do not exclude zeros throughout its source ball. The following local statement is enough to define its derivatives at zero when a logarithm is used.

Lemma 7.2 (A local logarithm)**.** Let H be holomorphic on every finite-dimensional restriction of a complex norm ball $\|W\| < R$, with compatible values on overlaps. Suppose $|H(W)| \le M$ there and $H(0) \ge m > 0$, where $M \ge m$. On the original ball,

$$|H(W)/H(0) - 1| \le (1 + M/H(0))\|W\|/R.$$

Moreover H has no zero on

$$\|W\| < \frac{mR}{M+m}.$$

The logarithm of $H(W)/H(0)$ defined to be zero at the origin is holomorphic on every finite-dimensional restriction of this ball. On the smaller ball $\|W\| < mR/(2M+m)$, its absolute value is at most $\log 2$. ■

Proof. Schwarz's lemma applied to $H - H(0)$ on each radial disc gives the first estimate, since its absolute bound is $M + H(0)$. For $W \ne 0$, apply Cauchy's coefficient bound to the one-variable function $\zeta \mapsto H(\zeta W/\|W\|)$ on $|\zeta| < R$. Summing its Taylor series away from the constant term gives

$$|H(W) - H(0)| \le \frac{M\|W\|}{R - \|W\|} < m.$$

Thus $|H(W)/H(0) - 1| < 1$, proving nonvanishing. The logarithm series converges locally uniformly on each finite-dimensional restriction, and defines compatible functions there. The value at zero is prescribed by the same series. On the smaller ball the displayed difference is less than $m/2$, so the ratio differs from one by less than $1/2$. The absolute sum of its logarithm series is then at most $\sum_{k\ge1} 2^{-k}/k = \log 2$. □

Corollary 7.3 (Finite-volume residual cumulants)**.** Let $c_i > 0$, and let h be holomorphic on $\sum_i c_i|z_i| < 1$, with $h(0) = 1$ and $|h(z)| \le Q$, where $Q \ge 1$. If $\mathcal{E}$ is holomorphic near zero, $\Gamma(z) = e^{\mathcal{E}(z)}h(z)$ and $\mathcal{E}(0) = 0$, then

$$|\partial_1 \cdots \partial_n(\log \Gamma - \mathcal{E})(0)| \le (\log 2)\,[n(2Q+1)]^n \prod_i c_i.$$

The logarithms are the branches equal to zero at the origin. ■□

Proof. Apply lemma 7.2 with $m = 1$, $M = Q$ and $R = 1$ in the weighted parameter norm. In its smaller ball of radius $1/(2Q+1)$, $\log h$ is bounded by $\log 2$. Choose coordinate circles with radii tending upward to $1/[n(2Q+1)c_i]$. Their products remain inside that ball before the limit. Multivariable Cauchy gives the product of the inverse radii. Each variable is differentiated only once. □

For a normalized sourced partition function satisfying $|H(W)| \le e^{C_+\ell^4}$ and $H(0) \ge e^{-C_-\ell^4}$, the radius supplied by this lemma may decay exponentially in the volume. No uniform global logarithm is asserted. The common cell normalizers used below are instead built from the local source energies themselves; their construction does not take a logarithm of the full partition function.

Lemma 7.4 (Separated derivatives of local energies)**.** Let $a_s = L^{-s}$ for $0 \le s \le K$. Suppose a family of domains X has an integer size $m(X) \ge 0$ and source dependence region X^+ with $\operatorname{diam} X^+ \le C_0 a_s(m(X)+1)$. Let $e_{s,X}(\sum_i z_i f_i)$ be holomorphic on $\sum_i |z_i|\|f_i\|_\sharp < r$, with bound $Ae^{-\kappa m(X)}$, and depend only on the source in X^+. Assume that the number of size-m domains whose dependence region meets $\operatorname{supp} f_1$ is at most $Ha_s^{-4}e^{\kappa_D m}$, where $\kappa > \kappa_D$. Assume the domain sums converge normally on smaller source balls at each fixed cutoff. For $n \ge 2$ and pairwise support distance at least $d > 0$,

$$\left|\partial_1 \cdots \partial_n \sum_{s=0}^{K} \sum_X e_{s,X}\left(\sum_i z_i f_i\right)\Big|_{z=0}\right| \le CAH \max\{1, d^{-4}\}(n/r)^n \prod_i \|f_i\|_\sharp. \tag{7.v}$$

The constant depends only on L, C_0 and the entropy gap, and is independent of K. ■

Proof. A zero test gives a zero derivative, so assume all test norms are positive. Locality makes the mixed derivative vanish unless X^+ meets every support. A contributing domain therefore has $m(X) \ge d/(C_0 a_s) - 1$. Cauchy circles with radii approaching $r/(n\|f_i\|_\sharp)$ bound its derivative by $A(n/r)^n e^{-\kappa m(X)} \prod_i \|f_i\|_\sharp$. The count through the first support and the strict entropy gap bound the sum at scale s by the same prefactor times $CHa_s^{-4}e^{-cd/a_s}$.

For $0 < d \le 1$, the scales with $a_s \ge d$ form a geometric sum bounded by Cd^{-4}. At smaller scales write $a_s \asymp dL^{-j}$; their sum is at most $Cd^{-4}\sum_{j\ge 0} L^{4j}e^{-c'L^j}$, which converges. For $d > 1$, replacing d by one only increases the sum. Normal convergence justifies termwise differentiation at each cutoff, and these estimates give its uniform bound. □

A finite-volume root count may take $H = C\ell^4$. A count anchored to a fixed radius-R support may instead take $H = C(R+1)^4$. The same proof covers both uses. A point counterterm $U(W) = \sum_p u_p(W(x(p)))$ has zero mixed derivative when the supports are disjoint, since no point reads two of the variables. Together with corollary 7.3, lemma 7.4 gives a cutoff-independent separated cumulant bound whenever Q has such a bound at the chosen finite volume. This remains a finite-volume statement if Q depends on ℓ. A comparison of successive cutoffs is sufficient under the following additional hypotheses.

Proposition 7.5 (A sufficient comparison of two cutoffs)**.** Fix the physical torus and real tests $f_1, \dots, f_n$ whose supports have mutual distance on the torus at least $d > 0$, where $n \geq 2$. Use the same coefficients $z = (z_1, \dots, z_n)$ for these tests at every cutoff, on one fixed open polydisc about zero inside all the admitted source balls. Write

$$\Gamma_K(z) = \frac{Z_K(\sum_i z_i f_i)}{Z_K(0)} = e^{\mathcal{E}_K(z)} h_K(z), \qquad h_K(0) = 1.$$

Assume $\mathcal{E}_K$ consists of the local source energies and possibly point terms, as in lemma 7.4. Use uniform locality, decay and rooted counts on that polydisc at the physical scales $a_s = L^{-s}$, including a strict entropy margin. Suppose the two energies at cutoffs K and $K+1$ can be paired for all indices at $s \leq K/2$, padding absent terms by zero, using domains with the same complete source dependence regions and the same rooted count. Assume that for each paired index the difference is bounded by

$$C(1+s)\epsilon_K e^{-\kappa m(X)},$$

with the same strict entropy margin. On the common polydisc suppose $|h_K - 1|, |h_{K+1} - 1| \leq 1/2$ and $|h_{K+1} - h_K| \leq \epsilon_K$. If $\sum_K \epsilon_K < \infty$, the separated cumulants for this tuple converge along the whole cutoff sequence. ■

Proof. Point terms have zero mixed derivative because each sees at most one of the separated tests. For a paired domain the mixed derivative of its difference vanishes unless its complete source read meets every support. Its size therefore satisfies $m(X) \geq d/(C_0 a_s) - 1$. Cauchy's estimate on a fixed smaller polydisc and the same rooted count as in lemma 7.4 bound the derivative of the coarse paired sum by

$$C(f_1, \dots, f_n)\epsilon_K \sum_{s \geq 0} (1+s) L^{4s} e^{-cdL^s}.$$

At $s > K/2$ bound the two local energies separately by the same individual estimates. The tail $\sum_{s > K/2} L^{4s} e^{-cdL^s}$ is summable in K, since interchanging the two sums gives at most $\sum_s (2s+1) L^{4s} e^{-cdL^s}$.

For the residuals, the segment between $h_K(z)$ and $h_{K+1}(z)$ stays in the closed disc of radius $1/2$ about one. Integrate $1/\zeta$ along it on the branch based at one to obtain $|\log h_{K+1} - \log h_K| \leq 2\epsilon_K$. Cauchy's formula bounds the mixed derivative by a constant times ϵ_K. The cumulant is the derivative of $\mathcal{E}_K + \log h_K$ at zero, so its consecutive differences are summable. The triangle inequality now gives convergence. The hypotheses concern paired energies of the actual varying source and the common residual disc. Individual bounds at the two cutoffs supply neither comparison. □

8. An upper bound for the sourced history sum

Fix the cutoff K, torus side ℓ, and a real admitted run. The physical spacing at level j is L^{j-K} and its number of sites is $N_j = \ell^4 L^{4(K-j)}$. At a stop $n \leq K$ let $E_{(n)}$ be its common vacuum number, with the sign convention of [4, Equation (2.iv)]. Write $Z(W)$ for the unnormalized bare integral with insertion $e^{\mathcal{O}_a(W)}$, where $a = L^{-K}$ and $x(p)$ is the centre of plaquette p. Every use of a source below is within the common ordinary analytic radius of the frozen construction. On the tori used for state transfer we take this radius to be $2r > 0$ in the physical norm of the next section.

Write $\mathcal{E}_{(n)}(W)$ for the common source number at this stop. It consists of the local energies of Proposition 7.1, and any separately filed point counterterm $U(W)$, holomorphic with $U(0) = 0$. In that convention point terms are counted outside the local energies. The native filing has $U = 0$.

The quantitative inputs for the sourced sum concern the same positive reference laws and the same record spaces as the real run. The complete sourced densities can take complex values. We use the following local conditions on these densities.

(1) The exact frozen pushforward gives complex label measures $\tau_{\mathcal{S}}(W)$ with total $Z(W)$ after summing their masses. The product coordinates and their Borel domains are the actual ones. Their source-independent reference factors and the disjointness or normalized counting of hidden records are as in the zero-source history sum. Write $e^{v_{\mathcal{S}}(W)}$ for the numerical factor on label $\mathcal{S}$. Use the actual scalar assignment of eq. 7.iii, with the missing shares charged at their original levels. If point terms are separately filed, their missing shares must have the same per-address bound. The charged addresses are in the live record or owned by a retained operation which has remade them. As in the loaded history estimate of [4, Proposition 8.5], the weight $v(\omega)$ of every record compatible with the visible label covers the own-scale sites assigned to it, with bounded multiplicity.

(2) For the source-dependent changes of the remaining regular, remainder and boundary exponent, take the sum of absolute bounds on the actual carriers before summing the histories. On a final-site read the rooted bound is at most $A_0(1+\log x_n)$, including the contributions of all earlier levels after conversion to final-site units. For example, a sufficient set of separate origin bounds at age h is $a_h L^{-4h}$ per origin site with $\sum_{h\geq 0} a_h \leq C(1+\log x_n)$. On the other reads the rooted bound is $A_1(1+\log x_j)$ per charged site at its original level j. These bounds apply on the same admitted real stored variables on which the sourced terms are evaluated. A read owned by a retained operation is charged to its stored sites. This input includes the terms attached to an island boundary and the factors carried outside a drop-out.

(3) The actual retained operations satisfy the complete-read and fresh amplitude bounds of Lemma 5.7, including the collar constants and the kept-response corrections. The resulting source-exponent changes obey $|\Psi_\omega(W)| \leq A_2 n_\omega$ on the allowed exterior and stored reads. The real weighted masses with their exterior suprema inside the integrals are available at the fixed increased reserve weights used below, chosen before the large inverse-coupling floor. Each allocated failure factor on its real support has an absolute majorant e^{-b} times a declared residual reference factor, with a positive uniform constant in the paying gain b. The needed real retained mass bounds include those residual factors and cuts. At the increased reserve they retain the claimed order of every mark gain spent in that mass. Otherwise an assigned last-test share must supply that gain separately. The remaining Wilson factor has nonnegative real plaquette coefficients after the share allocation. Coefficients read on different plaquette fields use their own allocations. The change Ψ_ω concerns the remaining factors after the allocated shares and the separately charged local changes have been removed.

These are estimates on local formation, allocation and retained integrals. No sum over the final supports is among them. For the ordinary old-origin regular part of the second input, eq. 5.vi gives a bounded cost per final window when its actual reads satisfy that proposition's hypotheses. Covering the final sites by a bounded number of such windows per site gives the corresponding part of the final-site cost. Boundary and retained reads still use their stated local bounds. For an action weight at one fixed plaquette field, ordered counterterm profiles give the convex combination in the real history argument of [4, Equation (2.vi)]. The same linear identity applies to the real parts of complex weights. For example, if all running weights on that plaquette differ from their positive real reference by at most a quarter of that reference, its real action coefficient is at least three quarters of the coefficient obtained from the reference weights. A positive allocation has to be checked on the same carrier before spending its gain.

Proposition 8.1 (Ordinary complex-source stability)**.** Use the actual frozen source representation and the three local inputs above. Assume the unsourced history geometry, real cost, record and paying-event estimates of the upper history bound in [4, Proposition 8.4], with actual rounded running-minimum radii of exponent $r_{\mathrm{cell}} \geq 1$ and paying gains of order $(\log \theta_j)^p$ for $p > 6r_{\mathrm{cell}}$. The frame has $\theta_j \geq \theta_n - c_*$ and $x_j \asymp \theta_j$ for $j \leq n$. Then, above a fixed inverse-coupling floor,

$$|Z(W)| \leq \exp\{E_{(n)} + \operatorname{Re} \mathcal{E}_{(n)}(W) + C(1 + \log x_n)N_n\}. \tag{8.i}$$

Under the actual empty-label assumptions of the common-normalized partition bounds in [4, Theorem E], also

$$Z(0) \geq \exp\{E_{(n)} - C(1 + \log x_n)N_n\}. \tag{8.ii}$$

The constants are uniform in cutoff, volume, admitted frame and source, with the displayed data fixed. At a terminal first-passage datum $n = K$ with x_K in the fixed window about g^{-2}, the error terms in both estimates are at most $C_g \ell^4$. ∎

Proof. Factor out $e^{E_{(n)}+\mathcal{E}_{(n)}(W)}$ from the sum of the complex label masses. At zero the difference between the label number and $E_{(n)}$ is the real vacuum price of the history bound. For the sourced part, subtract the empty-label shares from the shares in the label at hand. At each missing or changed root the absolute source price is at most the convergent per-root sum of Proposition 7.1. A change of a share can cost twice that bound. With the local multiplicities this adds at most a constant per charged site at its original level. The second input adds at most $A_0(1 + \log x_n)N_n$ on final-site reads and changes the own-scale charge constant on other reads.

For example, a spent energy share Q can pay a gain b if the failure forces $Q \geq 4b$ and the source perturbation of that share satisfies $\operatorname{Re} \Delta \leq Q/4$. On that support,

$$1_{\{\text{failure}\}}|e^{-Q+\Delta}| \leq 1_{\{\text{failure}\}}e^{-3Q/4} \leq e^{-b}e^{-Q/2}.$$

The residual reference factor is then $e^{-Q/2}$ with the cut. Any reference mass cost of retaining the reduced energy belongs to its local mass bound. The source changes and the reassigned prices of a retained operation are bounded by $cv(\omega)$ for a fixed c, since $1 + \log x_j \geq 1$ in the weights on stored sites. Increase the reserve by that c while keeping the same restrictions on its visible records. This comparison takes place on the stored variables, before the exterior supremum and integration. Apply the product majorant of [4, Lemma 8.1] to the disjoint integration groups. It allows coupling of their exterior arguments without exchanging their suprema with their integrals. Use the same normalized hidden-record reference or the same disjoint record sum as in the real estimate. A hidden failure already used in a retained mass contributes no second gain.

For a final component C, let v_C be its charged volume and G_C the sum of its surviving paying gains. The resulting bound has a common factor $e^{C(1+\log x_n)N_n}$ and a factor $e^{Av_C - G_C}$ per component record. The own-scale genealogy charge of [4, Lemma 8.2] bounds v_C by a fixed constant times $\sum_e M^4 \check{R}^6_{j(e)}$, summed over these same events. The volume v_C already counts $1+\log x_j$ per own-scale site. The running-minimum comparison gives $1 + \log x_j \leq C\check{R}_j$ and $\check{R}_j \leq C(1 + \log \theta_j)^{r_{\mathrm{cell}}}$. Since $p > 6r_{\mathrm{cell}}$, increasing the floor makes half of G_C pay this charge with any fixed additional entropy constant. The other half leaves a gain at least $c(1 + \log \theta_n)^p$, because every nonempty component has a paying event and its level j satisfies $\theta_j \geq \theta_n - c_*$. In particular, for a fixed sufficiently large λ and the rooted code constant c_{e} of the real estimate,

$$e^{Av_C - G_C} \leq e^{-c(1+\log \theta_n)^p - \lambda|C|}e^{-c_{\mathrm{e}}v_C}.$$

The sum over records of $e^{-c_{\mathrm{e}}v_C}$ is at most one. Discard the remaining positive event gain when summing the partition function. It has not been converted to a probability. On a final graph of degree at most D_0, at most $D_0^{2(m-1)}$ connected sets of size m are rooted at a given cell, by traversing a spanning tree twice. Thus the sum of $e^{-\lambda|C|}$ over all possible final components is at most CN_n for $\lambda > 2 \log \max(2, D_0)$. Bounding the collection sum by the product of $1 + e^{-\lambda|C|}$ over

these components gives at most e^{CN_n}. This proves eq. 8.i. Apply the zero-source lower bound of [4, Theorem E] for eq. 8.ii. Here $\mathcal{E}_{(n)}(0) = 0$, so its vacuum factor is exactly $E_{(n)}$. At $n = K$, $N_K = \ell^4$, and the first-passage window bounds $1 + \log x_K$ at fixed g. □

Corollary 8.2 (Finite-torus separated cumulants)**.** On a compatible torus of side $\ell \geq 1$, assume the ordinary source upper bound and zero-source floor with errors $C_+\ell^4$ and $C_-\ell^4$, respectively, uniformly on the source ball of radius $2r$. Assume the domain terms of their common source number $\mathcal{E}$ obey the local-energy estimates of proposition 7.1, with the actual cubical indexing and strict entropy margin of lemma 6.1. A separate holomorphic point counterterm vanishing at zero is allowed in this same source number. For $q \geq 2$, let $f_1, \dots, f_q$ be real Lipschitz torus tests whose supports have mutual torus distance at least $d > 0$. Their joint cumulant under the finite Wilson probability satisfies

$$|\kappa_q^{\mu}(\mathcal{O}_a(f_1), \dots, \mathcal{O}_a(f_q))| \leq C_q(d) e^{qc_V\ell^4} \prod_i \|f_i\|_\sharp.$$

The constants are uniform in cutoff and support positions when the input constants are uniform. No support-radius bound is required, and c_V is independent of q. ■□

Proof. If any test is zero, the cumulant is zero. Assume all their norms are positive. Write $\Gamma = Z(W)/Z(0)$ and $h = e^{-\mathcal{E}(W)}\Gamma$. The two partition bounds give $h(0) = 1$ and $|h| \leq Q = e^{(C_+ + C_-)\ell^4}$. Increase the nonnegative cost constants if necessary. Apply corollary 7.3 with $c_i = \|f_i\|_\sharp / r$. It bounds the mixed derivative of $\log h$ by $C_q Q^q \prod_i \|f_i\|_\sharp$. For the derivative of $\mathcal{E}$, use lemma 7.4 with the global root count $H = C\ell^4$. Its contribution is at most $C_q\ell^4 \max(1, d^{-4}) \prod_i \|f_i\|_\sharp$. Point terms contribute zero to this mixed derivative by separation. The cumulant is the sum of these two derivatives. The inequality $\ell^4 \leq e^{\ell^4}$ absorbs the polynomial volume factor; one may take $c_V = C_+ + C_- + 1$ after adjusting $C_q(d)$. □

9. Transfer at fixed spacing

Fix a cutoff–coupling datum $a = L^{-K}, \beta$ and a cofinal admitted family of tori of side $\ell = 2L^m$. Hold a, β fixed in the volume limit. Assume ordinary source bounds of the form eq. 8.i and eq. 8.ii, with errors $C_+\ell^4$ and $C_-\ell^4$. The constants can be uniform over a class of data satisfying the same source estimates. The preceding terminal first-passage construction is one sufficient case when its local hypotheses hold.

Write V_ℓ for the real vacuum number in these bounds, and write their common source number as

$$\mathcal{E}_\ell(W) = U(W) + \sum_{s=0}^{K} \sum_{D} e_{s,D}(W), \qquad a_s = L^{-s}.$$

A separately filed point term $U(W) = \sum_p u_p(W(x(p)))$ uses holomorphic functions vanishing at zero and locally bounded uniformly in p, ℓ at each fixed cutoff. Each domain term vanishes at zero, is holomorphic and depends on W only through its values on D^+. Assume $|e_{s,D}| \leq Ae^{-\kappa m(D)}$ and $\operatorname{diam} D^+ \leq C_0 a_s(1 + m(D))$, with the same nonnegative integer size $m(D)$, and assume normal convergence on smaller source balls. The count needed for ordinary transfer is

$$\sum_{\substack{D^+ \cap B_R \neq \varnothing \\ m(D) \geq t}} e^{-\kappa m(D)} \leq C(R+1)^4 a_s^{-4} e^{-ct} \qquad (t \geq 0) \tag{9.i}$$

uniformly in every radius-R ball B_R and the torus. The rooted domain count of lemma 6.1 implies eq. 9.i because $a_s \leq 1$. Its finer dependence on R/a_s and its global estimate will be used in the graded argument.

Use the physical test norm $\|\cdot\|_\sharp$ of eq. 2.ii at $s = 0$. The ordinary source bound is used for all finite representations $W = \sum_i z_i f_i$ on the torus with $\sum_i |z_i| \|f_i\|_\sharp < 2r$. Constant sources can be included among these real profiles. Every local term depends on the function restricted to its read

region. In particular its value does not change under a different admitted list representing that same restriction. The transfer uses eq. 8.i and eq. 8.ii with constants C_+ and C_- per unit volume, respectively.

Fix a board of side $l = L^b \geq a$ for an integer b, with faces bisecting links, as in [2, Theorem C]. Let $p_{l,\ell}(F)$ be its finite chessboard seminorm. It takes the N^{-1} power of the expectation of the disseminated product, where $N = (\ell/l)^4$ and all copies of odd reflection parity are conjugated. The expectation in this definition is for the original positive Wilson measure at zero source. Suppose real test functions f_i assigned to a particular cell have supports in m_C balls of radius R, at distance at least $d_{\mathrm{f}} > a$ from its faces. Put $w_C = \sum_i z_i f_i$. The plaquette terms in $\mathcal{O}_a(w_C)$ use only bonds in the open cell. In every dissemination, the even and odd copy profiles of each f_i are real and vanish near the faces. Their Lipschitz constants on the whole torus are at most $\operatorname{Lip} f_i$. Indeed the copies have that bound on each cell, agree at the faces, and the bound on a geodesic follows by adding its cell segments. Their suprema are also bounded by $\|f_i\|_\infty$. Hence the disseminated source W^{ch} has an ordinary representation of radius at most $2\sum_i |z_i| \|f_i\|_\sharp$. This represents it with the two lists z_i and $\bar{z}_i$ evaluated after the ordinary analytic functions have been defined.

Lemma 9.1 (A holomorphic cell normalization)**.** With these hypotheses and $\sum_i |z_i| \|f_i\|_\sharp < r$, there is a holomorphic function $E_{C,\ell}(z)$ with $E_{C,\ell}(0) = 0$ such that

$$p_{l,\ell}(e^{\mathcal{O}_a(w_C)}) \leq \exp\{\operatorname{Re} E_{C,\ell}(z) + B_C\}, \qquad B_C = (C_+ + C_-) l^4 + C m_C (R+1)^4 \sum_{s \geq 0} a_s^{-4} e^{-c d_{\mathrm{f}}/a_s}. \tag{9.ii}$$

One may replace the second term in B_C by any uniform bound on the absolute discarded source numbers per board cell. At fixed cutoff the functions $E_{C,\ell}$ are locally bounded uniformly in ℓ. Along a cofinal subsequence they converge locally uniformly to a holomorphic E_C. For finitely many chosen cell profiles one subsequence suffices and depends only on the finite-torus source numbers and the profiles. ■

Proof. The disseminated product is $e^{\mathcal{O}_a(W^{\mathrm{ch}})}$, since the plaquette action is invariant under orientation reversal. Its expectation is the nonnegative real ratio $Z(W^{\mathrm{ch}})/Z(0)$, by finite chessboard positivity. The source bounds therefore control the Nth root of its modulus.

In the common source number keep the point terms and the domain terms with their complete read regions inside one open cell. Each kept term can be evaluated on the source copy in that cell alone. The kept terms in even cells are holomorphic in z. An odd cell term has the form $g(\bar{z})$ for a holomorphic g. Use its conjugate $\overline{g(\bar{z})}$ when forming the average. This is holomorphic by its local power series, and its real part is that of the actual term. Divide the resulting sum by N to define $E_{C,\ell}$. The sum vanishes at zero.

A discarded term is zero if it reads no nonzero source value. Otherwise its read region meets a support ball and is not contained in that ball's cell. Its diameter is at least d_{f}, even on the periodic quotient. Thus $m(D) \geq d_{\mathrm{f}}/(C_0 a_s) - 1$. Use eq. 9.i once for each of the $N m_C$ ball copies. The exponential estimate for $e_{s,D}$ yields the second term of eq. 9.ii after division by N. No nonzero point term is discarded. The series is finite because $a_s^{-1} = L^s$. The extensive part of the partition bound divided by N is $(C_+ + C_-) l^4$, which proves the estimate.

At fixed cutoff the point terms per cell are bounded on smaller source balls by their fixed-cutoff bounds times $C l^4 a^{-4}$. For the kept domain terms use eq. 9.i with $t = 0$ and the finite sum over $s \leq K$. This gives local bounds independent of ℓ on each smaller source ball. Montel's theorem gives a cofinal subsequence, and a successive subsequence choice works for finitely many cells. The limit has the same zero value. □

Theorem A (One source normalization for all periodic states)**.** Assume the ordinary upper and zero-source lower bounds on the cofinal tori, the local source numbers, diameter bound and tail estimate eq. 9.i above, and fixed-spacing chessboard domination of [2, Theorem C]. Let $f_1, \dots, f_q$ be real Lipschitz tests with supports in radius-R balls in physical space. For this list and datum there is a holomorphic function $E(z)$ on $\sum_i |z_i| \|f_i\|_\sharp < r$, with $E(0) = 0$, such that for every periodic Wilson DLR state ρ at the datum,

$$\left| e^{-E(z)} \rho \left(e^{\sum_i z_i \mathcal{O}_a(f_i)} \right) \right| \le Q_{q,R}. \tag{9.iii}$$

The same E serves all such states. The constant $Q_{q,R}$ is uniform in cutoff, states and support positions. For one test f of radius R and any two states ρ, ν,

$$|\rho(\mathcal{O}_a(f)) - \nu(\mathcal{O}_a(f))| \le C(R+1)^4 \|f\|_\sharp. \tag{9.iv}$$

■□

Proof. Omit zero tests. Fix $d_{\mathrm{f}} > 1$. In each coordinate the forbidden board offsets for one support ball are contained in an interval of length $2(R + d_{\mathrm{f}})$ modulo the side, with a rounding cost of at most $2a$ on the link-bisecting grid. Choose the allowed side l to be the first power of L exceeding $4q(R + d_{\mathrm{f}} + 1)$. The forbidden offsets cannot cover the available grid in any coordinate. This puts every ball at distance at least d_{f} from its cell faces and uses $l \le 4Lq(R+d_{\mathrm{f}}+1)$. The torus sequence eventually admits that board with even counts.

Group the tests by their cells and apply Lemma 9.1 to each occupied cell. Use one common subsequence and put $E = \sum_C E_C$. At fixed cutoff the cell observables are bounded functions of finitely many bonds, and their exponentials are bounded complex cell functions at every fixed z. Chessboard domination holds for every periodic DLR state along every compatible cofinal sequence. Applying it along this one gives eq. 9.iii, with $Q_{q,R} = \exp(\sum_C B_C)$ bounded using at most q cells and the side bound. The subsequence was chosen before using any state. This argument does not select the state as a torus weak limit.

For one test put

$$h_\rho(z) = e^{-E(z)} \rho(e^{z\mathcal{O}_a(f)}), \qquad h_\rho(0) = 1.$$

Holomorphy of the expectation follows from boundedness of the observable at fixed spacing. Its derivative is $h'_\rho(0) = \rho(\mathcal{O}_a(f)) - E'(0)$. Cauchy's estimate gives $|h'_\rho(0)| \le Q_{1,R}\|f\|_\sharp / r$. Subtract the corresponding identity for ν. To obtain the stated radius dependence, use a smooth partition of unity by integer translates, supported in fixed-radius balls with uniformly bounded first derivatives. Such a partition is obtained by dividing a nonnegative compactly supported bump, positive on a unit cube, by the positive periodic sum of its translates. At most $C(R+1)^4$ pieces meet the support of f. Each piece times f has $\sharp$ norm at most $C\|f\|_\sharp$ and fixed support radius, so summing the fixed-radius mean bounds proves eq. 9.iv. The derivative of the common normalizer itself can grow with the cutoff. □

Corollary 9.2 (Separated cumulants)**.** Under Theorem A, let $q \geq 2$. If the supports have pairwise distance at least $d > 0$, then for every periodic state at the datum,

$$|\kappa_q^\rho(\mathcal{O}_a(f_1), \dots, \mathcal{O}_a(f_q))| \leq C_q(d, R) \prod_{i=1}^{q} \|f_i\|_\sharp.$$

For arbitrary supports, the difference of these cumulants in two states has a bound with a constant depending on q, R alone. The constants are cutoff independent. For any two such states ρ, ν and pairwise separated tests, the mixed moment centered at ν obeys

$$\left| \rho \prod_{i=1}^{q} (\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))) \right| \leq \left(\sum_{\pi} \prod_{I \in \pi} b_{|I|} \right) \prod_i \|f_i\|_\sharp,$$

where $b_1 = C(R+1)^4$, $b_m = C_m(d, R)$ for $m \geq 2$, and the sum runs over the set partitions of the indices. ■□

Proof. Write $h_\rho(z)$ for the holomorphic function inside the modulus in eq. 9.iii. Apply corollary 7.3 to h_ρ, with $c_i = \|f_i\|_\sharp / r$ and $Q = Q_{q,R}$. Zero tests can be omitted. It bounds the mixed derivative of $\log h_\rho$ by

$$(\log 2)\, [q(2Q_{q,R} + 1)/r]^q \prod_i \|f_i\|_\sharp.$$

The common derivative of E cancels in a difference of cumulants of two states. Applying this bound to each state gives the overlapping-support assertion with a cutoff-independent constant.

For one state, it remains to bound the mixed derivative of E. If several cells are occupied, each cell summand lacks at least one variable. If only one cell is occupied, point terms have zero mixed derivative by separation. A contributing retained domain must see every support, so its diameter is at least d. Its size is at least $d/(C_0 a_s) - 1$. Cauchy's formula with one circle for each of the q source variables gives the per-domain bound $A(q/r)^q e^{-\kappa m(D)} \prod_i \|f_i\|_\sharp$. Apply eq. 9.i through the first support ball in each copy, and divide by the number of copies. The scale sum is $\sum_s a_s^{-4} e^{-c' d/a_s} \leq C \max(1, d^{-4})$, by the split at $a_s = d$ used in the proof of lemma 7.4. Thus the mixed derivative of $E_{C,\ell}$ is bounded by

$$C(R+1)^4 \max(1, d^{-4}) (q/r)^q \prod_i \|f_i\|_\sharp.$$

Local uniform convergence passes it to E_C. Adding the log bound proves the cumulant estimate. The moment formula sums products of cumulants over set partitions. In a singleton block use eq. 9.iv. All other cumulants are unchanged by the constant shifts, so the preceding bound applies to each of those blocks. □

Corollary 9.3 (Local differences between states)**.** Use Theorem A and the lattice translation invariance of the periodic states. Let $a \leq d \leq d_0$, with $d_0 > 0$ fixed and sufficiently small, and let the supports of $f_1, \dots, f_q$ lie in one ball of radius $5d$. Write D_q for the difference of their order-q cumulants in two periodic states at the datum, viewed as a multilinear form of the tests. For $q = 1$ use the difference of the means. Then

$$|D_q(f_1, \dots, f_q)| \leq C_q d^4 \prod_i \|f_i\|_\sharp.$$

If $\operatorname{Lip} f_i \leq \|f_i\|_\infty / (\vartheta d)$ for a fixed $\vartheta > 0$, the right side is at most $C_{q,\vartheta} d^{4-q} \prod_i \|f_i\|_\infty$. ■□

Proof. Use lattice shifts in a grid of spacing $a\lceil 12d/a \rceil$, restricted to a fixed cube of side $1/2$. Their number P is at least cd^{-4}. The translated balls are disjoint and fit in a single unit ball about the original centre for small d_0. For each argument $i < q$ take independent signs ϵ_v^i on these shifts, and put $\epsilon_v^q = \prod_{i<q} \epsilon_v^i$. Write $F_i = \sum_v \epsilon_v^i f_i(\cdot - v)$. At most one copy is nonzero at any point. For points in different copy balls, split their connecting segment at the two ball boundaries, where the

copies vanish. The lengths of the two inner segments sum to at most the full distance. This gives $\|F_i\|_\sharp \le \|f_i\|_\sharp$.

In the multilinear expansion, averaging the signs retains precisely the terms with all shifts equal, since the average is $\prod_{i<q} \mathbf{1}_{v_i=v_q}$. Simultaneous lattice translation leaves both states' cumulants invariant. Thus the average is $PD_q(f_1,\dots,f_q)$. For $q=1$ the same formula holds with all signs one. Apply the overlapping-support bound of the preceding corollary in the unit ball, or eq. 9.iv when $q=1$, and divide by P. The final assertion follows from the definition of $\|\cdot\|_\sharp$ and $d \le 1$. □

10. Separated moments with a fixed factorial exponent

Lemma 10.1 (One-cell cost across scales)**.** Assume the ordinary partition bounds, source locality and tail estimate of Lemma 9.1. Assume that this source system has the actual cubical domain indexing and complete fixed-layer reads of Lemma 6.1. Require $|e_{s,D}(W)| \le Ae^{-\kappa m_s(D)}$ for its actual tree length, with $\kappa > \kappa_D$. Let a board have side $l \le 1$. Let a real test v be supported in a ball of radius δ inside one cell, at distance at least δ from the faces, with $16a \le \delta \le l/4$. There are constants Q_0, p independent of the cutoff, cell and torus such that

$$p_{l,\ell}\Big(e^{z\mathcal{O}_a(v)-E_{v,\ell}(z)}\Big) \le Q_0(l/\delta)^p, \qquad |z|\|v\|_\sharp < r. \tag{10.i}$$

The normalization is the one in Lemma 9.1. It has the same fixed-cutoff locally uniform subsequential limits. ■

Proof. All contributing plaquettes use bonds of the open cell, by the distance to its faces. The reflected source is admitted in the ordinary radius $2r$, using the real reflected profiles in even and odd cells and the coefficients z and $\bar z$, respectively. Each profile vanishes near the cell faces. The common source number is split into kept and discarded terms as in Lemma 9.1. A nonzero discarded term sees a source ball and its complete read is not contained in that ball's cell. Thus $m_s(X) \ge c_0\delta/a_s - C_0$.

At scales $a_s \ge l$, bound all terms by the global sum in eq. 6.i and divide by $(\ell/l)^4$. This costs $C(l/a_s)^4$ per level. At $\delta \le a_s < l$, sum the ball count over the source copies and divide by their number. The cost per level is at most C, and there are at most $1+\log_L(l/\delta)$ levels. At $a_s < \delta$ the rooted tail gives cost at most $C(1+\delta/a_s)^4 e^{-c\delta/a_s}$. The coarse and fine costs have uniformly bounded sums. For the fine sum, δ/a_s grows by L each step and the polynomial can be absorbed in half the exponential. The partition cost is $Cl^4 \le C$, and every nonzero point term is kept. Taking the exponential of the resulting bound $C(1+\log(l/\delta))$ gives the assertion. □

Theorem B (Graded separated moments)**.** Assume the ordinary source estimates and actual rooted domain indexing of Lemma 10.1. Let $f_1,\dots,f_n$ be Lipschitz tests, each supported in a ball of radius R. For $n \ge 2$ assume that their supports have mutual distance at least $d > 0$. Set $\bar d = \min(d,1)$, with $\bar d = 1$ for $n=1$. Let ρ,ν be any two periodic Wilson Gibbs states at the same bare datum. Then there are a fixed integer k and $B(R) = C(R+1)^4$ such that

$$\left|\rho\left(\prod_{i=1}^n [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))]\right)\right| \le B(R)^n \bar d^{-5n} (n!)^k \prod_{i=1}^n \|f_i\|_\sharp. \tag{10.ii}$$

For $n \ge 2$ the joint cumulant of the observables under ρ has the same bound with $2B(R)$ in place of $B(R)$. The estimates also hold if both expectations are the Wilson expectation on a compatible admitted finite torus, with distances measured on that torus. The constants are uniform in n, cutoff, volume, state and support positions. They may depend on the fixed source estimates and geometry. ■□

Proof. We first take real tests. Choose a power of L for the cell side, $\bar d/(16L) < l \le \bar d/16$, and put $\delta = l/(64n)$. If $a > \delta/16$, counting plaquette centres gives

$$\|\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))\|_\infty \le C(R+1)^4 a^{-4}\|f_i\|_\sharp \le C(R+1)^4\delta^{-4}\|f_i\|_\sharp.$$

The product of these bounds is within the estimate proved below.

Now let $a \le \delta/16$. Use a smooth partition of unity on a grid of step $\delta/8$, and write $f_i = \sum_\gamma v_{i,\gamma}$. One construction uses nonnegative translates of a fixed bump and divides by their sum, which is uniformly bounded below. The bumps have bounded overlap, support in radius-δ balls and Lipschitz constant C/δ after normalization. The grid tiles a compatible torus since ℓ/l and $l/(\delta/8)$ are integers. There are at most $C(1+R/\delta)^4$ nonzero pieces per test, and $\|v_{i,\gamma}\|_\sharp \le C\delta^{-1}\|f_i\|_\sharp$.

For any choice v_i of one piece per test, choose a common board offset. In each coordinate there are l/a offsets of link faces. For each ball at most $4\delta/a+2$ of them put some face within distance δ of the ball. Since $n(4\delta/a+2) < l/a$, an offset remains in each coordinate. The selected balls are inside their cells with the required reserve. Nonzero pieces from different tests lie in different cells, since their nonzero supports are at distance at least d and a cell has diameter $2l < d$.

For each chosen v_i use Lemma 10.1 with $Q_i = Q_0(64n)^p$ and radius $r_i = r/\|v_i\|_\sharp$. Zero pieces can be dropped. At fixed cutoff choose one cofinal subsequence for these finitely many cell normalizations, and write E_i for the limits. For every subset $J \subset \{1,\dots,n\}$, chessboard domination [2, Theorem C] gives

$$\left|\rho\left(\prod_{i\in J} e^{z_i\mathcal{O}_a(v_i)-E_i(z_i)}\right)\right| \le \prod_{i\in J} Q_i, \qquad |z_i| < r_i.$$

This follows by applying the bound to the unnormalized cell functions at fixed z, then taking the limits of the scalar normalizations. In finite volume use the finite chessboard inequality and $E_{v_i,\ell}$ directly. The variables in this product are independent complex coordinates. Cauchy's formula, once per coordinate, bounds the product moment centered at $E_i'(0)$ by $\prod_{i\in J} Q_i/r_i$. Applied in state ν with one coordinate it gives $|\nu(\mathcal{O}_a(v_i)) - E_i'(0)| \le Q_i/r_i$. Expanding the product about these constants over all subsets proves the bound $2^n \prod_i Q_i/r_i$ with the requested centering.

Sum the last estimate over piece choices. The cost from the pieces of each test is at most $C(R+1)^4\delta^{-5}$ times its norm, so the bound is

$$[C(R+1)^4]^n n^{(p+5)n}\bar{d}^{-5n}\prod_i \|f_i\|_\sharp.$$

It includes the coarse-cutoff case. Since $n! \ge (n/e)^n$, the claim holds with $k = \lceil p+5\rceil$ and an adjusted constant. Complex tests follow by real and imaginary parts and absorbing 2^n.

Cumulants of order $n \ge 2$ do not change when constants are subtracted. Use the moment formula on each block of a partition. For block sizes $b_1,\dots,b_m$ one has $\prod_j b_j! \le n!$. After removing a factor $(n!)^{k-1}$, order the blocks and sum over their positive sizes. At fixed ordered sizes there are $n!/\prod_j b_j!$ ordered partitions. Hence the remaining sum, including the coefficient $(m-1)!$, is

$$n!\sum_{m=1}^{n}\frac{1}{m}\binom{n-1}{m-1} \le 2^{n-1}n!.$$

This proves the cumulant bound. □

For a version with a separation factor independent of n, apply $x^n \le e^x n!$ with $x = \bar{d}^{-5}$ in eq. 10.ii. This replaces $(n!)^k$ by $(n!)^{k+1}$ and the separation factor by $e^{\bar{d}^{-5}}$.

11. Uniform insertion in the reflected norm

Write τ_v for translation by v, so the support moves by v, and θ_t for time reflection at $x_4 = t$. At fixed spacing a, choose a cell C of side $l = L^j \ge a$ centered at a lattice site, with lower time face t. The ratio l/a is odd. Its faces are link mirrors and its middle plane is a site mirror. For a periodic Wilson Gibbs state ρ, the form

$$\langle B, A\rangle_t = \rho(\overline{\theta_t B}A), \qquad \|A\|_t^2 = \langle A, A\rangle_t \tag{11.i}$$

is positive semidefinite by the all-mirror theorem [2, Theorem B]. Its raw algebra consists of bounded Borel functions of finitely many bonds whose endpoints lie on the positive side of the link plane.

Lemma 11.1 (Cell multiplication and a time shift)**.** For a bounded complex cell function D using bonds with both endpoints in C, and any cofinal sequence of compatible tori, set $q_D = \limsup_\ell p_{l,\ell}(D)$. On the algebra of eq. 11.i, the map $T_D A = D\tau_{le_4} A$ satisfies

$$\|T_D A\|_t \le q_D \|A\|_t, \qquad |\langle B, T_D A\rangle_t| \le q_D \|B\|_t \|A\|_t. \tag{11.ii}$$

It therefore descends to the quotient by the null space. ■

Proof. Put $D^* = \overline{\theta_{t+l/2} D}$. The reflection identities $\theta_t \tau_{le_4} = \tau_{-le_4}\theta_t$ and $\overline{\theta_t D^*} = \tau_{-le_4} D$, followed by translation invariance of ρ, give $\langle B, T_D A\rangle_t = \langle T_{D^*} B, A\rangle_t$. These are identities on the raw algebra. For $K_D = T_{D^*} T_D$ and $m_b = \langle A, K_D^b A\rangle_t$, they imply

$$m_{2s} = \|K_D^s A\|_t^2, \qquad m_{2s+1} = \|T_D K_D^s A\|_t^2.$$

Cauchy–Schwarz gives $m_b^2 \le m_{b-1} m_{b+1}$ for $b \ge 1$. For $b = 2s+1$ use $K_D^s A, K_D^{s+1} A$. For $b = 2s \ge 2$ use $T_D K_D^{s-1} A, T_D K_D^s A$. If $m_0 = 0$, Cauchy–Schwarz gives $m_1 = 0$. Otherwise $m_1^N \le m_0^{N-1} m_N$. In the nonzero case this follows by iterating the increasing ratios m_b/m_{b-1}.

In m_N there are the two endpoint functions $\overline{\theta_t A}$ and $\tau_{2Nle_4} A$, with $2N$ alternating copies of D^*, D between them. Choose a coarse side $l' = L^u l$ large enough to contain the finite spatial and time support of A in one coarse cell above t. Center the spatial directions of that cell as in C and set its lower time face at t. Its faces are fine-board faces because L^u is odd. Choose N a multiple of L^u. The two endpoints are then in separate coarse cells outside the column. All used bonds of a constituent fine cell lie in the open coarse cell containing it.

Apply fixed-spacing product domination on this coarse board [2, Theorem C], taking volume to infinity for the fixed N. The two endpoint seminorms are at most $\|A\|_\infty$. For each middle cell the fine-to-coarse estimate in that theorem bounds its seminorm by the product of those of its fine factors. Translations changing the reference cell preserve the fine seminorm. Reflection through the middle of C permutes the fine board and conjugates its copy reflections, so invariance of the torus expectation and conjugation invariance give $p_{l,\ell}(D^*) = p_{l,\ell}(D)$. Since $q_D \le \|D\|_\infty$, for every $\epsilon > 0$ this proves $m_N \le \|A\|_\infty^2 (q_D + \epsilon)^{2N}$. Take Nth roots, send N to infinity along the indicated multiples, then let ϵ tend to zero. This includes $q_D = 0$ and gives the first estimate in eq. 11.ii. The second is Cauchy–Schwarz. □

Theorem C (A centered lattice insertion)**.** Assume the ordinary source estimates and actual rooted domain indexing of Lemma 10.1. Let $a \le l \le 1$ be an allowed cell side and h a real Lipschitz test supported in the radius-$l/4$ ball about its center. For any two periodic Wilson Gibbs states ρ, ν at the same bare datum,

$$\begin{aligned}\Big|\rho\Big(\overline{\theta_t B}[\mathcal{O}_a(h) - \nu(\mathcal{O}_a(h))]\tau_{le_4} A\Big)\Big| \\ \le C\|h\|_\sharp \|B\|_t \|A\|_t.\end{aligned} \tag{11.iii}$$

The constant is uniform in the states, cutoff, cell side and the functions A, B in the raw algebra. ■□

Proof. For $a \le l/64$, Lemma 10.1 with $\delta = l/4$ yields one-variable normalizers with

$$p_{l,\ell}(e^{z\mathcal{O}_a(h) - E_{h,\ell}(z)}) \le Q, \qquad |z|\|h\|_\sharp < r,$$

with fixed Q. If $3 \le l/a < 64$, every contributing plaquette is internal to C. Its vertices in each coordinate are at distance at most $l/4 + a/2 < l/2$ from the cell center. The number of plaquettes is bounded by $C64^4$. Take $E_{h,\ell} = 0$ in this case and enlarge Q. At $l/a = 1$ the ball contains no plaquette centre, so the same bound holds.

Take a cofinal subsequence at fixed cutoff with normalizer limit E_h. For each z in the disc, homogeneity and convergence imply $q_{D_z} \le Q$ for $D_z = e^{z\mathcal{O}_a(h) - E_h(z)}$. Apply eq. 11.ii to D_z.

Cauchy's formula applies to the expectation because the fixed-cutoff observable is bounded and E_h is holomorphic. The derivative at zero is bounded by $(Q/r)\|h\|_\sharp\|B\|_t\|A\|_t$ and inserts $\mathcal{O}_a(h)-E_h'(0)$. With $A = B = 1$ in state ν the same argument bounds $|\nu(\mathcal{O}_a(h)) - E_h'(0)|$ by $(Q/r)\|h\|_\sharp$. The $D = 1$ case of eq. 11.ii bounds the resulting scalar term. The same normalizer works in both states. If $\|h\|_\sharp = 0$ the insertion vanishes directly. □

Proposition 11.2 (Insertion for already convergent separated moments)**.** Assume the hypotheses of Theorem C uniformly along $a_j \downarrow 0$. Suppose that at each datum ρ_j, ν_j are two periodic Wilson Gibbs states and the centered moments converge on every fixed separated tuple of compactly supported C^1 tests. Denote the limits by σ, write $\phi(f)$ for an entry with test f, and set $\sigma(1) = 1$. Let F, G be finite linear combinations of monomials whose entries have mutually separated supports within each monomial and are compactly supported C^1 tests lying strictly in positive time. The form $\langle G, F\rangle_\sigma = \sigma(\theta_0\overline{G}\,F)$ is positive semidefinite. For $l \le 1$ an allowed power of L and real $h \in C_c^1$ supported in the radius-$l/4$ ball about $(y, 0)$,

$$|\sigma(\theta_0\overline{G}\,\phi(\tau_{(l/2)e_4}h)\tau_{le_4}F)| \le C\|h\|_\sharp\|G\|_\sigma\|F\|_\sigma. \tag{11.iv}$$

Time translations by ue_4 for $u \ge 0$ act by contractions in this seminorm. Both operations respect null vectors. ■

Proof. Multilinear telescoping and Theorem B give equicontinuity on each fixed separated tuple when each test is translated by a vector tending to zero. Indeed the support of a difference is in the union of two nearby supports, the gaps stay positive, and the difference tends to zero in the sharp norm for compact C^1 tests. This also proves translation invariance of the limits, by approximating a fixed real shift with lattice shifts and using the invariance of both states at each datum.

Let y_j be a nearest spatial lattice point to y and put $v_j = (y_j - y, a_j/2)$ and $t_j = a_j/2$. The point $(y_j, t_j + l/2)$ is a lattice site for all sufficiently fine cutoffs. Translate F, G by v_j and the inserted test by $v_j + (l/2)e_4$. The equality $\theta_{t_j}\tau_{v_j} = \tau_{v_j}\theta_0$ identifies the norm tuples. The centering state ν_j is invariant under this lattice mirror by [2, Theorem B], so its reflected centering constants agree as well. For the lattice form apply eq. 11.i to the polynomials of centered lattice observables. All supports are in the indicated open half-spaces when j is large, since the tuple gaps are fixed and each plaquette has diameter $O(a_j)$. Positivity of the limiting form follows. Apply eq. 11.iii to the same polynomials and take the limit using the fixed-gap equicontinuity. This gives eq. 11.iv with its stated constant even when a norm vanishes.

At fixed cutoff and $u \ge 0$ a lattice time, invariance and reflection Cauchy–Schwarz give

$$\|\tau_{ue_4}A\|_t^2 \le \|A\|_t\|\tau_{2ue_4}A\|_t.$$

If $\|A\|_t = 0$ this gives the assertion directly. Otherwise iterate, using $\|\tau_{2^k ue_4}A\|_t \le \|A\|_\infty$, and send k to infinity. For any nonnegative real u, take nonnegative lattice times tending to u and use the same fixed-gap passage. The norm bound for the insertion itself follows by using the polynomial $G = \phi(\tau_{(l/2)e_4}h)\tau_{le_4}F$ as the left argument of eq. 11.iv. This polynomial has separated entries. The two norm estimates give null-space descent. □

Corollary 11.3 (Near-mirror insertion)**.** Under Proposition 11.2, let $0 < \eta \le 1$ and $f \in C_c^1$ be supported in $\eta \le x_4 \le 3\eta$ with support diameter at most R. Then

$$\|\phi(f)\tau_{4\eta e_4}F\|_\sigma \le C(1 + R)^3\eta^{-4}\|f\|_{C^1}\|F\|_\sigma. \tag{11.v}$$

Fix $0 < c < 1/64$. If $k \in C_c^1$ is supported in the ball of radius $c\eta$ about y with $|y_4| \le \eta/8$, and the entries of F, G lie strictly above $\eta/4$, then

$$|\sigma(\theta_0\overline{F}\,\phi(k)\,G)| \le C\eta^{-4}\|k\|_\sharp\|\tau_{-(\eta/4)e_4}F\|_\sigma\|\tau_{-(\eta/4)e_4}G\|_\sigma. \tag{11.vi}$$

The constants are independent of the number of entries in the monomials and the number of monomials in F, G. ■□

Proof. Take the largest power l of L at most η and use a smooth partition at mesh comparable to l with pieces supported in balls of radius $l/4$. There are at most $C(1+R)^3\eta^{-3}$ nonzero pieces, since the support has time width 2η. The original test vanishes on the two time faces, so $\|f\|_\infty \le \eta\|\partial_4 f\|_\infty$. As $l > \eta/L$, the derivative of a partition factor costs at most a fixed multiple of $\|f\|_{C^1}$ in the sharp norm of the piece.

For a piece with center time b, let $t = b - l/2$. Translate its product with $\theta_0\overline{G}$ and $\tau_{4\eta e_4}F$ by $-te_4$. The left polynomial becomes the reflection of $\tau_{te_4}G$, and the right one is $\tau_{le_4}\tau_{(4\eta-t-l)e_4}F$. Both extra translations have times at least $\eta - 3l/4 > 0$ because $\eta - l/4 \le b \le 3\eta + l/4$. Apply eq. 11.iv and time contraction to each piece. The resulting bilinear estimate has cost $C(1+R)^3\eta^{-3}\|f\|_{C^1}\|G\|_\sigma\|F\|_\sigma$. Set $G = \phi(f)\tau_{4\eta e_4}F$ and use $\eta^{-3} \le \eta^{-4}$. Real and imaginary parts handle a complex f.

For eq. 11.vi, set $\xi = c\eta$ and translate the whole expectation in time by $2\xi - y_4$. The translated test lies in $[\xi, 3\xi]$ in time. The left polynomial is the reflection of $\tau_{(y_4-2\xi)e_4}F$, and the right polynomial is $\tau_{4\xi e_4}\tau_{(-y_4-2\xi)e_4}G$. The two polynomials before reflection and the 4ξ shift are positive translates of their $\eta/4$ downshifts, by at least $\eta/8 - 2\xi > 0$. Apply eq. 11.v, Cauchy–Schwarz and time contraction. The translated test has diameter at most $2c$, and the factor c^{-4} is fixed. This gives the asserted bound. □

12. A two-source estimate at running stops

The preceding transfer can be used at a finer first-passage stop than the unit-scale stop. The needed ordinary estimates must be available at each of these stops, with their own site counts.

Proposition 12.1 (Running-stop estimate for separated pairs)**.** Take first-passage data of [4] with terminal index K and reference floor h. Assume the self-calibrated prefix is common under the admitted changes of units. On the cofinal tori of every first-passage stop used below, assume the ordinary source bounds in the units of that stop, with $C_+ + C_- \le C(1 + \log x)$ at terminal weight x. The source radius, local energy bounds, entropy margin and geometric constants are uniform over these stops. The minimum allowed torus side may depend on the stop.

Fix $\vartheta > 0$. Suppose f_1, f_2 have support radii at most d, mutual support distance at least d, and $\operatorname{Lip} f_i \le \|f_i\|_\infty/(\vartheta d)$. There are $c, C > 0$ such that, for $0 < d \le c$ and every periodic Wilson state ρ at the original datum,

$$\left|\kappa_2^\rho(\mathcal{O}_a(f_1), \mathcal{O}_a(f_2))\right| \le C\big[1 + h + \log(1/d)\big]^C \|f_1\|_\infty \|f_2\|_\infty. \tag{12.i}$$

The constants are uniform in cutoff and support positions. The bound holds in fixed-gap limits of these states whenever the separated second cumulant converges. In particular multiplication of the right side by d^2 tends to zero as $d \downarrow 0$ at fixed h. ■

Proof. In the real trajectory, call i a record if $x_i < x_t$ for all $t < i$. The initial and first-passage indices are records. Let $J = 1 + \lceil c_5/\lambda_{\mathrm{s}} \rceil$, with the lower drift constants of [4, Theorem A and Proposition 4.4]. After a record i with $i + J \le K$, $x_i - x_{i+J} \ge \lambda_{\mathrm{s}} J - c_5 > 0$, so there is a new record within J steps. If fewer steps remain, K supplies the next one. Thus the depths of the records, listed from 0 to K, have gaps at most J.

Put $n = \lfloor \log_L(c/d) \rfloor$. For $K < n$, the original spacing is greater than d/c. A support of radius d then contains at most $C(1 + d/a)^4$ plaquette barycentres, and the elementary bound $0 \le a_p \le 2$ proves eq. 12.i. If $K \ge n$, choose the largest record depth $b \le n$. Then $cL^{-J-1} \le L^b d \le c$. For $b > 0$, the prefix ending at $K - b$ is a first-passage trajectory for the new floor $h_b = x_{K-b} - B_{\mathrm{s}}$, where B_{s} is the original upper step constant. Every earlier entry exceeds x_{K-b} by the record property. The old first-passage condition gives $h_b > h$. The upper drift and the terminal window give $x_{K-b} \le h + B_{\mathrm{s}}(b+1)$. For $b = 0$ keep the original stop.

Dilate by L^b. The bijection of the bare plaquettes preserves both Wilson conditional specifications and their scalar observables with tests $f_i^{[b]}(x) = f_i(L^{-b}x)$. Hence it carries periodic Gibbs states to periodic Gibbs states and preserves the cumulant. The common-prefix hypothesis identifies the

stopped normalization in the new units. Now the support radii are at most c, the distance is bounded below by cL^{-J-1}, and $\|f_i^{[b]}\|_\sharp \le C\|f_i\|_\infty$.

Apply theorem A in these units at the new stop. Its proof uses at most two occupied cells of bounded side. The cell normalizer bound of lemma 9.1 is the sum of $(C_+ + C_-)$ times that bounded cell volume and a uniform energy tail. Therefore the residual generating function has bound $Q \le \exp[C(1 + \log x_{K-b})]$ on a fixed source disc. The mixed derivative of the common holomorphic energy is bounded uniformly at the new separation floor by the local-energy estimate in corollary 9.2. For the residual H with $H(0) = 1$, its log derivative is $\partial_{12}H(0) - \partial_1 H(0)\partial_2 H(0)$. Cauchy gives a bound $C(Q + Q^2)\prod_i \|f_i^{[b]}\|_\sharp$. Inserting the upper bound on x_{K-b} proves eq. 12.i. The volume limit in the transfer is at this fixed spacing and stop, so a stop-dependent minimum torus side causes no loss. The fixed-gap passage follows by convergence of the cumulant. □

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